

Solve It!

Mysteries

for enquiring minds

2



Karl Jacklin
A G.A.T.E.WAYS Challenge



Quality ready-to-use
resources



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Mysteries

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Karl Jacklin

A G.A.T.E.WAYS Challenge



Karl Jacklin has been a teacher, school principal, Manager of Personnel and Administration in the North West Metropolitan Region of the Victorian Education Department, a teachers' college lecturer, tutor in statistics and measurement at Deakin University, and manager of his own software development and computer consultancy company. He has also presented many highly successful programs with G.A.T.E.WAYS (Gifted and Talented Education, Extension and Enrichment), an independent, non-funded organisation which aims to provide opportunities for children of 'like-minds' to participate in programs that will challenge them intellectually, and help develop their individual talents.

For further information: info@gateways.edu.au

G.A.T.E.WAYS directors, Win Smith and Jill Lawrence, wish to express their deep appreciation to Karl for his dedication in writing this exciting and challenging publication.

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ABN 074 266 023

108 Main Road

Clayton South VIC 3169

Ph: +61 3 9548 0193

www.blake.com.au

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Introduction



Solve it! Mysteries for enquiring minds is a collection of 10 hypothetical scenarios across a range of challenging topics. It is designed to encourage students to use problem-solving techniques and logical reasoning, and to consolidate and test their comprehension and writing skills.

Five scenarios are contained in this book, and five more are in Book 1. Each one includes teacher notes which provide a background briefing, related activities and sample answers, and photocopiable information sheets which students will need to solve the mystery.

Each unit of work is suitable for individual or group work within the regular classroom, or team competitions and extension programs.



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How to use the mysteries

There are five mysteries in Book 2 and a further five in Book 1. Each of the 10 scenarios:

- 1 outlines an event requiring a logical explanation, based on the information in the student resource material. The final explanation may be definitive, such as *The murderer was ...*, or it may be the most appropriate of several plausible alternatives. Students should be encouraged to ask: *What does this evidence tell me? How does it logically lead to a solution? What other evidence might be useful?* Some students will arrive at the correct solutions, but not adequately explain how they reached their conclusions. They do need to be encouraged to provide evidence of a reasoned solution process
- 2 provide a basis for discussion about the mechanics of research, reading comprehension, clear expression and analytical and deductive reasoning. The scenarios are designed to enable students to put into practice such skills as:
 - sorting out the main issues
 - distinguishing between what is relevant and what are merely ‘red herrings’
 - identifying plausible alternatives against which to test one’s evidence
- 3 is self-contained, in that no additional information is required to reach a satisfactory conclusion
- 4 is open-ended and can be used by you and your students as a prompt for further research, exploration and discussion
- 5 can be used to initiate a range of associated class activities. For instance, students might debate a broader version of the topic or respond with creative writing or role plays
- 6 provides an opportunity to study and employ scientific, logical, and evidence-based methods of evaluating controversial issues
- 7 includes tasks that enable teachers to integrate knowledge and skills from several curriculum areas into the scenario.

SUGGESTED APPROACH

How you develop a topic is entirely up to you. Each of these 10 mysteries deals with an issue of interest that can be enjoyable in its own right. The mysteries can, however, also be the source of a lot of other intellectually stimulating and challenging activities involving research, debate, discussion, role play, mathematical analysis, scientific experimentation and creative writing.

Teachers who have used this material report that without extension activities, their classes have completed the tasks within an hour to an hour and a half.

Teacher’s notes pages

Each scenario includes suggestions as to how best approach the mystery, as well as more detailed descriptions about the skills and knowledge that are explored within it.

A set of sample answers to the questions posed on the student report is provided in the teacher’s notes for each mystery. The sample answers are not intended to be definitive, but do provide a suggested level of detail.

Student information sheets

Complete sets of photocopiable student pages need to be prepared before the lesson. Students should be encouraged to organise the data they obtain from these sheets, using devices such as tables, Venn diagrams or concept maps to help them logically consider and analyse all the available evidence.

CURRICULUM LINKS

The disappearing dinosaur

- **Science:** Fossils and organisms of the past
- **Personal development:** Values, motives and decision-making

Murder at 15 Parson Street

- **Science:** Analysing forensic evidence such as ballistics, blood, temperature, handwriting
- **Personal development:** Values, motives and decision-making

Industrial espionage

- **Maths:** Coding and encryption

The haunted house

- **Science:** Using science to solve problems

The party

- **Science:** Meteorology, global warming and climate change

See *Solve It! Mysteries for enquiring minds Book 1* for more information.



The disappearing dinosaur

TEACHER'S NOTES

In this mystery, a valuable fossil of a velociraptor (a small, meat-eating dinosaur) disappears from the local museum.

The questions

- Who did it?
- How was it done?

To unravel the mystery, students need to:

- describe the crime
- interview any witnesses
- analyse the collected evidence
- examine motives.

After taking all the material into account, students must arrive at a plausible solution.

- Is the evidence conclusive?
- How confident are the students that their solution to the crime will lead to a conviction?

Suggested approach

One helpful tool is to have the students conduct a mock trial involving each of these participants:

- a judge
- a prosecutor
- a defence lawyer
- the accused
- the various witnesses
- the police detective
- the courtroom spectators
- a jury comprised of students from another class unfamiliar with the details of the case.

The participants will need to familiarise themselves with their roles and carefully prepare responses. They should consider the following points.

- What is the precise charge against the accused?
- What are the responsibilities of the judge?
- How is the trial to be conducted?
- What evidence is admissible, and what isn't?
- What is perjury and how might it be recognised during the trial?
- To what extent does the judge direct the jury?
- How does the prosecutor present the case?
- What questions should be asked, and more importantly, what answers sought?
- If questions do not provoke the desired responses, how can they be rephrased?
- What is the role of the defence lawyer?
- How and under what circumstances does the defence lawyer examine witnesses?

Ideally, each participant should prepare a script in advance, although there should also be room for spontaneity.

DEBATE TOPIC

Museums are becoming less relevant in our rapidly changing technological society.

Background

The rate of change in our society is increasing to such an extent that museums will be unable to continue providing a reasonable historical overview. The volume of change in, say, the next 100 years may far exceed everything that has occurred previously. Decisions will therefore need to be made about what to include in such collections. However, this has the potential to give an erroneous or biased view of history. What is excluded may say just as much about us as what is included. Then there are the costs involved. Museums may end up becoming quaint collections of archaic and largely irrelevant materials.

RELATED ACTIVITIES

The brontosaurus

Get the students to research and prepare a short description of the 'brontosaurus'. Most will describe it as one of the best known of the giant sauropods. However, some will discover that in fact it was the most famous dinosaur that *never* lived! American palaeontologist Othniel Marsh discovered and named the apatosaurus, but later rediscovered a set of its fossilised bones that included some vertebrae and a skull from another large sauropod. He thought he had discovered a new species, which he named the brontosaurus. For decades, museums displayed full-sized replicas of the brontosaurus.

Palaeontology

Discuss the work of palaeontologists and how they unravel the mysteries of long extinct creatures by studying their fossilised bones.

Get the students to write an imaginary press release in which they describe a newly discovered dinosaur called the 'chickosaurus'. Have them refer to the fossil as a basis for their hypothetical description of its appearance, probable life style, and the period during which it existed.



The disappearing dinosaur

TEACHER'S NOTES

Extinction of the dinosaurs

In 1940, Walt Disney released an animated feature film entitled *Fantasia*. It was a collection of seven sequences, each on a different theme, and each underscored by a different piece of popular classical music. One of the sequences, set against Igor Stravinsky's *Rite of Spring*, dealt with the origin and development of life on Earth up to and including the age of the dinosaurs. It concluded with the demise of these creatures as Earth's climate gradually changed, becoming hotter and drier. However, this view of dinosaur extinction has been challenged. There have been at least seven different theories, the most recent being the mass extinction event that occurred 65 million years ago when a large comet or asteroid impacted with Earth.

This is an excellent research topic – it has depth and it's still controversial. Why did every single species of dinosaur (regardless of size) perish, yet other species of reptiles, birds, mammals and amphibians (frogs, crocodiles, lizards, etc.) survive? Even species of dinosaurs no bigger than chickens became extinct, despite the fact that the lizards on which they preyed did survive.

REFERENCES

Burnie, D, *The Kingfisher Illustrated Dinosaur Encyclopaedia*, Kingfisher Publications, London, 2001

Encyclopaedia of Dinosaurs and Prehistoric Life, Dorling Kindersley, London, 2001

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

Most likely motive for the crime (10 points)

It might have been Jeremy Bensen's intention to sell the extremely valuable velociraptor fossil on the black market, but a more powerful motive for the crime was to seek revenge on Frederick Debruce who Bensen believed had been unfairly promoted above him to the curator's position. It was probably also his intention to undermine, and possibly even replace, Debruce.

Most likely perpetrator of the crime (15 points)

Mr Jeremy Bensen had most to gain from the crime, and he had the knowledge and opportunity to carry it out successfully.

Reasons for your decision (15 points)

- 1 Jeremy Bensen was passed over for the job of museum curator and was perhaps facing the sack.
- 2 He had the knowledge and opportunity to carry out the crime.
- 3 He was missing from his office for the 90 minutes it would have taken him to dismantle and conceal the velociraptor fossil.
- 4 He made sure that he was the one with the responsibility for searching the area in which he had hidden the stolen fossil, and so ensured that it wasn't discovered.

Explanation of how the crime was carried out (5 points)

Jeremy Bensen dropped in on Frank Vella, the museum's Chief Security Officer, in his office at about 9.30 pm for a cup of coffee and a brief chat. During this time he put something in Vella's coffee that produced symptoms similar to a severe gastric upset. This ensured that Vella would be unlikely to carry out his normal inspection rounds of the building during the next few hours.

Bensen also knew that by this time the cleaning staff would probably have finished their nightly duties. He then went to the velociraptor exhibit and, during the next 90 minutes or so, carefully dismantled and packed the fossil. He then took it to the storage vaults and hid it where he knew it would be safe until he could remove it from the building without detection (probably packaged as some other exhibit).

It was while Bensen was doing this that Carmen Montana noticed that Bensen's office was in darkness. On his way out of the building at 11.30 pm, Bensen briefly spoke again to Vella. It was either during this time, or in the morning when he notified Vella of the theft of the fossil, that Bensen replaced the security tapes with blanks.

Bensen then notified the Chairperson of the Museum Board of the theft rather than his immediate boss, which added to Debruce's embarrassment over the affair. Previously, Bensen had stolen Brian Worthington's keys and had him sacked, to suggest him as a possible suspect.

Hiding place of the fossil (5 points)

In the museum storage vaults

YOUR TASK – You are a detective with the Carrington Police Department, investigating the theft of a valuable dinosaur fossil from the Carrington Museum during the evening of Tuesday, 7th October. Although insured, the loss of the fossil – a velociraptor worth \$2.5 million – is a huge embarrassment to the organisers, and may jeopardise future exchanges of such exhibits between museums. What makes the case unusual is that the fossil was moderately large (approximately 1.3 metres in length) and would have had to have been carefully dismantled by the thief before it was removed from the building. You suspect an inside job because none of the museum's alarms was triggered, and the thief chose the most valuable fossil from the potential targets. You have collected all the evidence and interviewed all of the potential suspects. Now your task is to determine the guilty party and discover the location of the fossil.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task and think about the sort of evidence that might help you determine the culprit, and the method and motive for the crime.
- 2 Make a list of what you consider to be the most plausible explanations.
- 3 Read the interview statements, consider the museum floorplan and the security log, and consider whether the information they provide favours a particular explanation. Do they suggest other possibilities?
- 4 State who you think committed the crime, give the motive and then explain how the crime was carried out and where the fossil was hidden.

The Carrington Clarion

Wednesday 8 October

'Dino' Disappears – Museum Mystified

by Rayna Naidoo

Police are investigating an audacious robbery at the Carrington Museum that occurred during Tuesday evening, when the valuable fossil of a 1.8-metre velociraptor was stolen from its display case. The velociraptor was a fast and ferocious dinosaur killer that lived on Earth during the late Cretaceous period some 68 million years ago. Its light body, muscular legs, sharp sickle-like claws, and habit of hunting in packs made it the perfect killing machine.

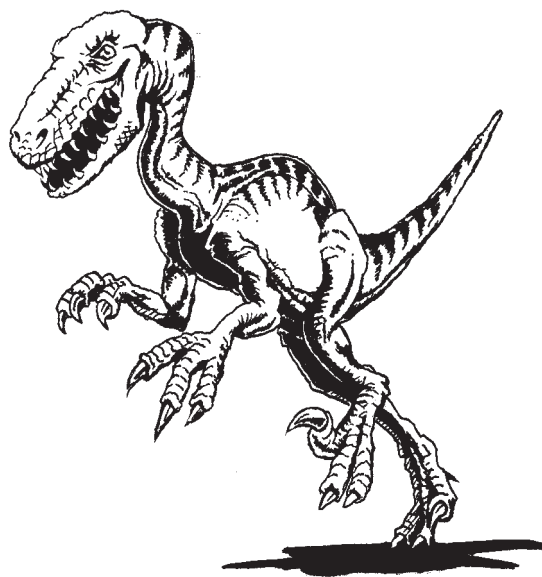
In more recent times, the velociraptor has enjoyed a surge in popularity, mostly due to the award-winning 1993 movie *Jurassic Park*, directed by Steven Spielberg.

The fossil, found in Mongolia in the 1920s, was valued at over \$2.5 million, and was on loan from

the Yangtze University of Natural Sciences.

The museum's curator, Frederick Debruce, believes that the fossil will be difficult to dispose of on the black market as there have been few such discoveries, and therefore any attempt to display or sell it will be highly conspicuous. Mr Debruce explained that the fossil must have been completely dismantled by the thief before removal from the museum. It would therefore have to be re-assembled prior to it being sold.

'Consequently, I believe that the thief would have needed expert knowledge of palaeontology in order to carry out such a difficult task,' Mr Debruce stated. A police spokesperson said enquiries were proceeding normally.



Velociraptor (speedy thief), was a member of the Dromeosaur family. It had a 12-centimetre-long, needle-sharp, retractable claw on each foot, and a large brain for its size, giving it intelligence and rapid reactions

Report by Mr Frederick Debruce, Museum Curator

The fossil was on loan from the Yangtze University of Natural Sciences, and was insured for \$2.5 million. In actual fact, it was priceless, as it was a perfect specimen and quite rare.

I think it would have taken the thief at least 90 minutes to disassemble the exhibit, and since it must be re-assembled before selling, I assume the thief has considerable knowledge of the creature's skeletal structure.

I can't explain why the museum's security system did not activate when the fossil was removed from its display case, as any change in weight on the floor of the case should have triggered the alarm. Similarly, the security cameras should also have recorded the theft, but the tapes are apparently blank. In any case, our security officer Frank Vella was supposed to have made a circuit of the museum every two hours, during which time he should have discovered the theft. He claims that he did check the raptor exhibit several times during the night, and that everything seemed normal.

Frank's been with us for 20 years, during which time we've had many valuable exhibits. I can't imagine he was involved in any way. It makes no sense. Bensen suggested that it might have been the maintenance fellow I sacked last week, or perhaps an accomplice.

I left the museum about 6 o'clock and went alone to a local Chinese restaurant, after which I went to the cinema. I got home about 11 pm, read for a while and then went to bed about midnight.

The first I knew of the theft was when Sir Martin Brown, the Chairman of the Museum Advisory Board, rang me this morning. It was a very embarrassing moment I can assure you. He was most upset about the theft of the velociraptor. I told him that it was an incident waiting to happen, and that it reinforces my decision to shake up the place and make it more modern. Unfortunately, I've met a bit of resistance from some of the longer-serving staff. But as I've said to them, it's a matter of shaping up or shipping out. We just have to move with the times. Fortunately the exhibit is insured, however, the whole episode has left me with egg on my face.



Report by Sir Martin Brown, Chairperson of the Museum Advisory Board

Yes, I did ring Debruce when I heard about the theft. I was most upset, as it will have all sorts of unpleasant repercussions for the Museum. I'm not all that certain that we did the right thing

in appointing Debruce ahead of Bensen, who has served us loyally and competently for 15 years. Maybe now the Board will realise that modernisation for its own sake is not a sound basis for change. If it's not broken, don't fix it, I say!

I can't understand why the museum's alarm system didn't work. We spent a considerable sum in getting it up to standard in preparation for this important exhibit. As I said to Debruce at the time, 'Wouldn't we be better served by increasing the number of security staff?' Unfortunately, he convinced me that this was the best way to go. It makes you wonder whether it was an inside job.



Report by Mr Terry Biggs, Security Officer

I come in at 6 am each day in advance of the normal arrival time of staff. My job is to turn off the security system and do a quick check of the building. Mr Benson is usually the first to arrive for work. He normally gets in between 6.30 am and 7.00 am. This morning he got here

just after I arrived. In fact it was Mr Benson who discovered the theft of the dinosaur. He was very upset, and got Frank back in straight away.





Report by Mr Frank Vella, Chief Security Officer

The first I knew about the robbery was when it was reported to me by Mr Jeremy Bensen. He's Mr Debruce's deputy. He said he'd been passing the raptor exhibit on his way to his office when he noticed, to his horror, that the display case was empty.

He asked to see my security log. That's the book in which I record the results of my two-hourly checks of the museum. Then he suggested that I ring both the police and Mr Debruce, and inform them of the robbery while he checked the security tapes. I was amazed when he told me that they were blank. Either the tapes were switched when I was on one of my tours of the building, or the system was faulty. Maybe the techs will be able to answer that one.

I wasn't really surprised that Mr Bensen discovered the robbery as he likes to wander around the museum before opening time just to make sure that everything's ship-shape. He's been on the staff here for 15 years, and was deputy to Mr Debruce's predecessor before Mr Debruce was appointed earlier this year. Prior to that I think he was a scientist of some sort, a paleo something or other. He takes great pride in the museum and was most upset by the robbery, which he thinks will harm our reputation.

As for the alarm system, that's got me really puzzled. It should have gone off as soon as the display case was opened, and certainly once the fossil was lifted from its stand. It has something to do with electronic sensors and changes in weight. Anyway, it didn't work. Not that I'm surprised. Mr Debruce sacked one of the maintenance staff only last week, and he might have sabotaged the system out of spite. I guess they'll fully test the system as soon as Forensics have finished their investigations.

Actually, I didn't get around to checking the building again after Mr Bensen left. We'd had a cup of coffee and a brief chat at about 9.30 pm, shortly after which I had a bit of a gastric upset. I think that the milk may have been off. From then on I wasn't able to go too far from the office area. Fortunately Mr Benson has his coffee black or he might have been crook too. I secured the building and went home at about 12.30 am, only to be called back by Terry at about 6.45 am. Mr Debruce was in a real state.

We have about 40 people on staff, and since Mr Debruce took over, all employees have had to provide a police clearance check. I guess it's because we are responsible for such valuable items.

I've worked at the museum longer than anyone else, just on 20 years in fact, but as a result of this robbery, I might be in for the big chop. I know that Mr Debruce is keen to outsource security as part of his plans to modernise the place.



Report by Mrs Carmen Montana, Albright Cleaning Service



The cleaning team comes on duty at about 6.00 pm each night. We work our way through every office and display room in the building except the storage vaults, and it normally takes us about four hours to clean the place thoroughly. I think we'd have noticed if one of the exhibits had been tampered with. I do remember seeing Frank Vella several times during the evening as he carried out his regular security checks of the building. Mr Bensen was also working back late, as he does most nights, but the lights were off in his office when I left the building at about 10 pm.

Report by Ms Claudine Morrell, Chief Accountant

I worked in my office until 9.15 pm, at which time I packed up and went home. I didn't realise the velociraptor fossil had been stolen until Frank told me about it the next morning. A loss like that is certainly going to damage the museum's reputation.



The disappearing dinosaur

6



Report by Mr Jeremy Bensen, Deputy Museum Curator



I discovered the theft of the velociraptor fossil when I looked in on the exhibition at about 6.45 am. Most of our staff commence work at or before 8.05 am, as we open to the public at 9 am. I like to get in early just to make certain that everything is ready to go by opening time, and to deal with any outstanding paperwork. Once we had a leaking water pipe, and one of our most valuable exhibits was nearly ruined. Fortunately I discovered it just in time. That's why I make it my business to keep a vigilant eye on things.

I've no idea how the velociraptor was removed from the building undetected, but I can say that things haven't been going too well of late. Nothing serious, just a few minor slip-ups. One of the maintenance staff was sacked last week for losing his keys. He claimed they'd been stolen from his desk drawer. At first Frederick was inclined to give him the benefit of the doubt, but I told him that it was just too big a security risk to keep the fellow on, so he sacked him. He might be your culprit.

I worked back quite late last night to get the preliminary planning done for the next big exhibition. We are focusing on different aspects of Antarctica – you know, exploration, wildlife, and the like. I'm very excited about it. By the time I finished it was about 11.30 pm, and after a few minutes talking to Frank I headed home, just before he started another security check of the building. I rang the Chairman of the Advisory board this morning to tell him about the Antarctic Exhibition and was surprised to discover that no one had let him know about the theft of the velociraptor. He was quite upset, as you'd expect.

Report by Ms Lillian Cheng, Secretary to Mr Debruce

I was Mr Debruce's secretary when he was in charge of the Northern Suburbs Library Complex. He turned eight old-fashioned standalone libraries into one modern interlinked, computer-based, information resource centre. It was his success in that job that got him his



current position, which was quite amazing. Once his position was confirmed, he sacked his predecessor's secretary and offered me the job. I jumped at it. I know he wants to shake this place to its foundations; bring it into the 21st century so to speak. He's already intimated to several long-serving senior employees that they're on borrowed time. I know that Mr Bensen and Mr Vella are on his 'hit' list. I think that Mr Debruce has already spoken to Mr Bensen about his future. I'm sure that in the long run it will all be to the benefit of the users of our museum.

Report by Mr Brian Worthington, ex-employee from Maintenance

I've worked at the museum for nearly 10 years. Then along comes Mr Debruce with all his fancy ideas, and I'm out. He claims that I lost my keys, and compromised the security of the whole complex. I told him that they had gone missing from my locked desk drawer while I repaired a broken window pane in Mr Bensen's office. I normally take my keys



with me, but Mr Bensen was insistent on having the window repaired immediately, and I just forgot them. I didn't give them another thought until Mr Debruce came storming into my office. It puzzled me that Mr Debruce knew that my keys had disappeared before I did. Mr Bensen agreed, and I was grateful when he said he'd take the issue up with Mr Debruce, but obviously he had no success. I know that those keys accessed every lock in the building, including the security system, but I assure you, I had nothing to do with their loss or the break-in.

The disappearing dinosaur



Report by Detective Sergeant Frank Mulloy, Investigating Officer, Carrington CID

I attended the crime scene at 9.15 am. There was no sign of forced entry to any section of the building.

I interviewed all relevant individuals (transcripts attached) and there are some quite definite inconsistencies that need to be followed up. Several of them have sufficient motive for the crime.

I had the forensics team fingerprint the room in which the fossil of the velociraptor was exhibited, and as you would expect at a public venue, there were a myriad of unidentifiable prints. However, one set of prints was found inside the glass display case. Mr Jeremy Bensen suggested that they might be his, as he had set up the exhibit. This turned out to be the case.

I am at a loss to explain how the fossil was removed from the crime scene. External cameras only showed staff leaving the building, and none of them was carrying anything large enough to conceal the fossilised remains of the raptor. Mr Debruce informs me that Mr Bensen coordinated a thorough search of the building, with no success.

I am sure that this is an inside job, but by whom, and for what purpose? Mr Debruce believes that the fossil would be difficult to sell. The failure of the museum's security system to activate during the robbery seems too coincidental not to be sabotage.

Further investigation into the backgrounds of the museum staff has revealed that the Chief Accountant, Ms Claudine Morrell, frequently visits the Casino, and currently has an outstanding gambling debt of \$9500.

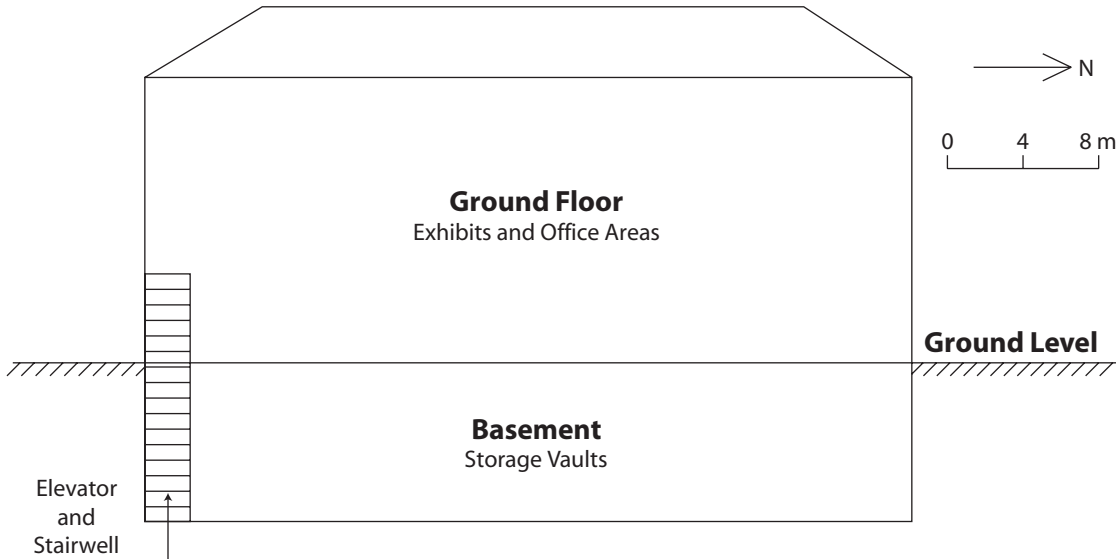
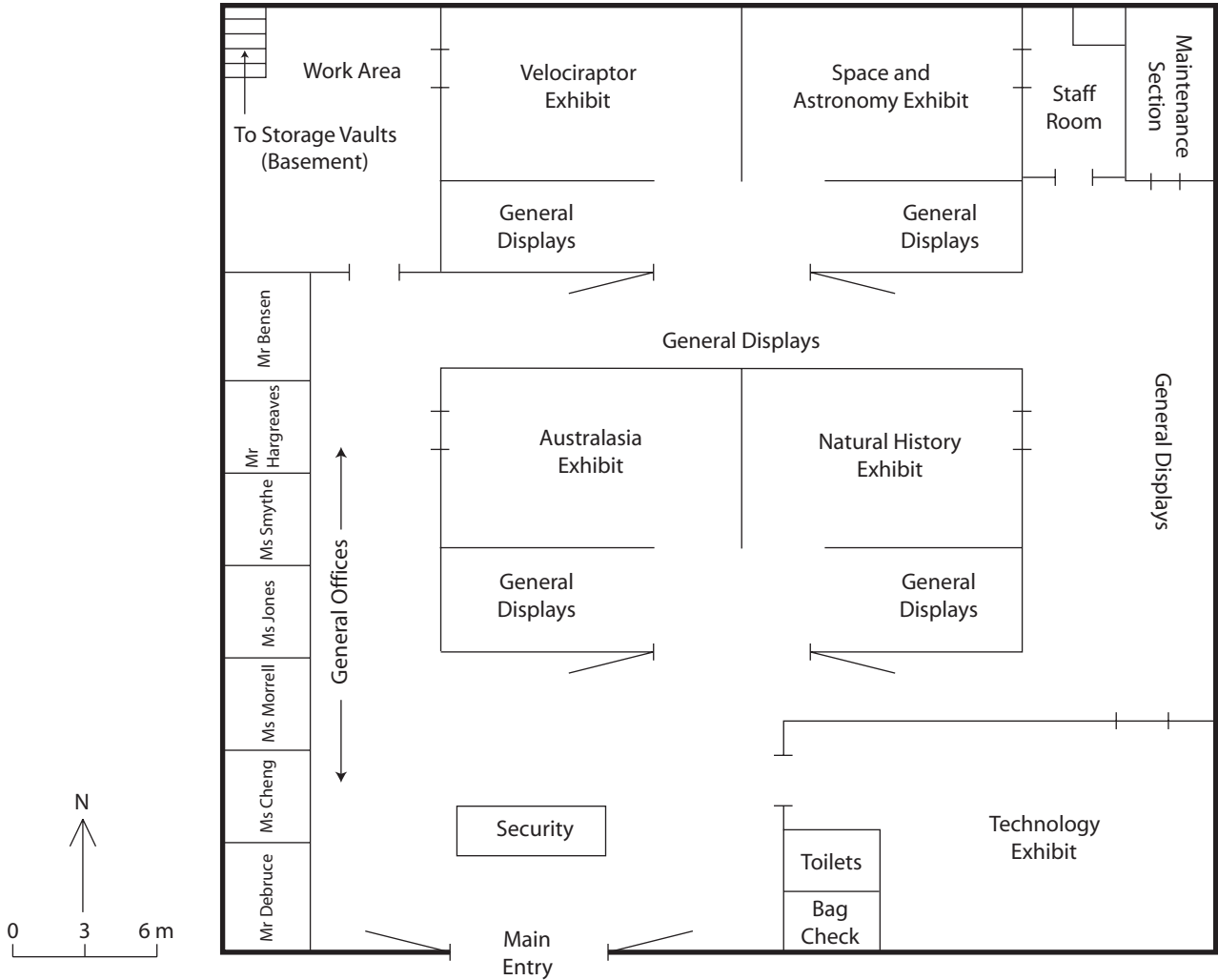


SECURITY LOG – 7TH OCTOBER 2006

Name	Position	Time in	Time out	Signature
Jeremy Bensen	Deputy Curator	6.45 am	11.35 pm	Jeremy Bensen
Frederick Debruce	Curator	8.45 am	6.00 pm	Debruce
Lillian Cheng	Secretary	8.45 am	5.45 pm	Lillian Cheng
Claudine Morrell	Chief Accountant	8.10 am	9.15 pm	CM
Carmen Montana	Cleaning Contractor	6.00 pm	10.06 pm	Carmen Montana
Frank Vella	Chief Security Officer	5.00 pm	12.30 am	FW Vella
Terry Biggs	Security Officer	6.00 am	12.10 pm	Terrance Biggs
Con Sammos	Security Officer	11.55 am	5.40 pm	Sammos
Jennifer Jones	Assistant Accountant	8.15 am	5.45 pm	Jam Jones
Cameron Jones	IT Manager	8.30 am	6.05 pm	C Jones
Angela Chrystostomou	IT Programmer	8.15 am	4.45 pm	Angela Chrystostomou
Marilyn Smythe	Admin Officer	8.25 am	5.45 pm	Smythe
Jack Hargreaves	Admin Officer	8.20 am	6.00 pm	J.P. Hargreaves
Vince Gerin	Exhibits Supervisor	7.45 am	5.05 pm	Nonconform
Karen Thompson	Exhibits Officer	8.25 am	5.45 pm	Karen Thompson



Museum Floor Plan



The disappearing dinosaur – report

Write on the back of this page if you need more space.



Most likely motive for the crime

Most likely perpetrator of the crime

Reasons for your decision

[illegible]

Explanation of how the crime was carried out

[illegible]

Hiding place of the fossil



Murder at 15 Parson Street

TEACHER'S NOTES

In this mystery, a murder has been committed in a city-centre office located in a vacant building.

The questions

- Who did it?
- Why?

Who did it?

This crime mystery is a little more complex than 'The Disappearing Dinosaur' because:

- the motive of the killer and the method by which the crime is carried out are not as straightforward
- there are the additional issues of alibis and opportunity
- there is forensic evidence to consider.

Why did the person do it?

The students, as investigating officers, must build an airtight case against the accused, based on a number of supporting pieces of evidence. Perhaps, when viewed alone, a piece of evidence might not be conclusive, but when added to others, the supposition of guilt becomes stronger.

DEBATE TOPIC 1

'An eye for an eye' is the only way to reduce the crime rate.

Background

Crime is a major concern for society. Trying to find solutions to crime often becomes an election issue. Candidates will promise to come down more heavily on criminals, make penalties harsher, increase the police force, improve detection through the increased use of modern technology, put rehabilitation programs in place, make prisons less degrading and brutalising, and so on. However, as this debate topic highlights, there is a point of view that sees punishment as a prime deterrent. 'The worse the crime, the harsher and tougher the punishment,' some suggest. But which is the more successful deterrent – the fear of punishment or the likelihood of detection?

DEBATE TOPIC 2

Work is a necessary distraction.

Background

Why, in general, do people work? Why do some people work when they don't need to? Why do others avoid work at every opportunity? Is work only a means of enabling us to do what we really want to do during the remainder of our time? Does work have an intrinsic value? What would society be like if nobody needed to work? What would the world be like if robots did all the work?

The connection between this second debate topic and the mystery is not as immediately straightforward as the first debate topic. It should be noted that none of the squatters have jobs. Is this by choice, or is it a situation that is largely beyond their control? The 'hippies' of the 1960s were a group who claimed to have opted out of what they called 'the rat race'. Other groups of people, right up to the present day, have made the same decision.

The mystery and this debate topic provide a good introduction to what has the potential to be a fascinating study of the place of work in our society. Over the centuries, the western world in particular has moved irrevocably from subsistence to specialisation, from manual labour to mechanisation – and many parts of the rest of the world are now moving in this direction too. Where once a person leaving school might have anticipated a job for life – or at least until retirement – the situation is changing to one where a school leaver can expect to change jobs at least four or five times during a working career. But what of those who drop out? Those whose skills are no longer relevant? Those who are considered too old to retrain?



Murder at 15 Parson Street

TEACHER'S NOTES

RELATED ACTIVITIES

Forensic science

The work of the police incorporates the latest in science and technology; fingerprinting, DNA testing, microscopy, chromatography, ballistics, and so on. There are many research and discussion activities as well as practical activities such as taking fingerprints and conducting matching exercises that can be developed around these topics.

The criminal investigation process

This activity highlights aspects of the general process of criminal investigation.

- 1 Students observe a video clip of a crime. This can be something that the teacher and the students have produced, or an appropriate scene from a movie or television.
- 2 Each student writes down a description of what they have seen. What will be of interest here is the expected variations in their accounts of the crime. There will undoubtedly be omissions, additions and distortions.
- 3 A directed interview then occurs. Each student responds in writing to a series of questions dictated by the teacher. These questions should be prepared in advance by the teacher, and based on the selected video clip.

For example:

- What did the criminal look like?
- Were they of tall, average or short height?
- What build were they – small, medium, heavy?
- What was their hair colour?
- What type of clothing was worn?
- Did the person have any distinguishing markings such as tattoos or scars?
- Was the person wearing any jewellery?
- Were there any other clearly discernible factors?
- Was there a motor vehicle and, if so, what type and colour?
- Did the criminal have an accomplice?

- 4 Students are then asked to draw the criminal.

This is mainly undertaken as a bit of fun, but it is interesting to again look at the variation that occurs. Of course, the police use Photofit. If the teacher has actually produced a video tape of the crime, then a digital camera can be used to photograph the faces of a number of volunteers. By using a suitable computer graphics program, these faces can be divided into their basic parts – eyes, mouth, hair, ears, chin, head shape, and so on. It's a time-consuming exercise, but the original file of facial features can be used again. The students then describe these facial features, and a computer composite is built up.

- 5 This information forms the basis of the police investigation. The police detectives will have:

- a description of the criminal
- a description of the crime.

- 6 Now, the students write down the various steps as they imagine they might be undertaken by the detectives investigating the case. This may include:

- collecting and collating any evidence found at the scene of the crime
- checking to see whether there have been any other similar crimes carried out in the area, and whether there are any known criminals who have been guilty of such crimes in the past (looking at *modus operandi*)
- checking with known informants who might provide useful information about the crime
- checking available databases for information, including matches for fingerprints, DNA, physical descriptions, car registration, etc.
- building up a list of suspects
- interviewing suspects
- checking alibis
- narrowing the field of suspects
- re-interviewing the prime suspect/s
- arranging a line-up
- laying charges
- prosecuting the case.



Murder at 15 Parson Street

TEACHER'S NOTES

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

An initial list of suspects and their possible motives (if any) for the crime (5 points)

- **Peter Davis** (Squatter): Admits to being occasionally bad-tempered and can be violent, but has no apparent motive for murdering Groover.
- **Greg Jones** (Squatter): Has no obvious violent tendencies, and no apparent reason for murdering Groover.
- **Julie Symonds** (Squatter): Is generally friendly, especially towards Groover, and has no apparent motive for the murder.
- **Henry Katz** (Squatter): An extremely violent and jealous disposition, especially towards Groover – a seemingly strong motive for the murder.
- **John Atkin** (Property owner): Has a violent dislike of squatters, and has the motive and opportunity and seems to be prepared to use a gun if necessary.

The name of the murderer (5 points)

Julie Symonds

Evidence on which your decision is based (20 points)

- 1 A ballistics test matched the rifling on the bullet removed from the deceased, Mr Stanton, with the bullet fired from the gun found in the church grounds.
- 2 The fingerprint on the murder weapon matched the one taken from Ms Julie Symonds's middle finger on her right hand.
- 3 Although a gunshot was heard in the vicinity of the murder scene (15 Parson Street) at approximately 2.30 pm, a forensics examination of the body at 3.30 pm revealed that the actual time of death was approximately four hours earlier, around 11.30 am. This assessment was based on a 3.3°C fall in body temperature at an average cooling rate of 0.83°C per hour.
- 4 Julie Symonds had no alibi for this period. In her statement to the police, she lied about leaving at 1 pm for the cinema with Katz and

suggesting that they had been together up until this time. In fact, Katz left the squat at about 10.15 am and did not meet up with Julie Symonds again until she arrived at the cinema at 1 pm.

- 5 At approximately 11 am, John Atkin, the property owner, overheard a girl at what he later discovered was the murder scene, arguing heatedly with someone.
- 6 In her police statement, Julie Symonds referred to Grant Stanton by his given name, whereas the other squatters knew him only as Groover. Since she claimed that she had not been alone with him at the squat prior to the murder, either she had lied, or they had known each other from an earlier time.
- 7 Peter Davis, one of the squatters, stated that Julie had been friendly towards Groover. However, Katz stated that she had told him she found Groover's attention unwelcome.
- 8 Julie Symonds was absent from her seat at the cinema during a period that overlapped the time that witnesses heard a gunshot in Parson Street (2.30 pm).
- 9 Ahmet Massoud stated that a woman in a distraught condition had boarded his bus at about 2.40 pm (coinciding with the time he first heard the police sirens) and that she told him that a man had been murdered. The only person who could have been aware of this fact at that time was the murderer. Mr Massoud told police that the woman alighted from the bus at the stop adjacent to the Bijou Cinema.
- 10 The witnesses thought that it had been a man in dark clothing running from the murder scene at around 2.30 pm. The police line-up showed Julie Symonds to be a physically big woman so, from a distance, and wearing dark loose fitting clothing, such as a tracksuit and beanie, she could quite easily have been mistaken for a man.
- 11 The handwriting on the torn-up note found at the scene of the crime suggested that the person who had written it was being blackmailed. The handwriting was later matched to Julie Symonds.



Murder at 15 Parson Street

TEACHER'S NOTES

Method and motive (20 points)

Julie Symonds had previously known – and had probably had a relationship with – Grant Stanton (known only to the others as Groover). When they met again by chance at the squat, Julie was frightened that Katz would become jealous if he found out about this earlier friendship. Katz was well known for his violent nature.

However, instead of keeping their secret, Stanton threatened to tell Katz unless Julie paid him \$200. So Julie decided to scare him off. At first she tried to make Katz angry by pretending that Groover was forcing his attentions on her. Although this angered Katz, it only resulted in a couple of heated arguments. Consequently, Julie felt that she had no alternative but to pay Stanton the money, but the day after she had paid Stanton the \$200, he demanded more. This was probably the argument overheard by John Atkin, the property owner.

Whether she had already considered a fallback plan in the event that Stanton became greedy, or whether it was just an unpremeditated act, it was then that Julie Symonds killed him. The gun could have belonged to any of the squatters, but its small size suggested that it might have been Julie's. She fired the shot through a pillow to muffle the sound, after which she replaced the spent cartridge.

The next thing that Julie had to do was get out of the squat as soon as possible in case anyone turned up unexpectedly, although she knew that Peter and Greg had gone job hunting for the day, and she'd already arranged to meet Katz at the cinema at 1 pm. She quickly searched through Stanton's belongings in the hope of finding the \$200 she had given him the day before, but it wasn't there. What she did find, however, was a black tracksuit and beanie.

When they were seated in the cinema Julie offered Katz a sip of her drink, to which she'd added a strong sedative. He was already tired, and so it didn't take long for the drug to work. In the meantime, Julie created a commotion in the theatre by putting her feet on the back of the seat in front just as the usher passed by. She knew the usher would remember her should the police check up on her story.

What she didn't anticipate was that the usher would also notice that she was missing from her seat at about 2 pm. She later claimed that she had gone to the cinema restroom, but the usher's statement seriously weakened her alibi. In fact, she'd left the cinema, returned to the squat, put on the black tracksuit, beanie and dark glasses, fired a shot into the floor, moved the filing cabinet to cover it, and rushed from the building, darting up Parson Street into South Street, and then into an alleyway where she dumped the clothing.

She then ran to the bus stop on the corner of Parson Street and Valley Road where she caught the bus, getting off opposite the cinema in Loftus Road, returning to her seat in time to see the remainder of the film. On the way back to the squat, she dropped the gun into some bushes in the grounds of the local Methodist church. Perhaps she intended to retrieve it later.

Her purpose in all of this was to mislead the police into thinking that the murder had occurred at 2.30 pm, at which time she believed she had a sound alibi. Unwittingly, she had overlooked the forensic evidence, the observant usher, and the discovery of the murder weapon on which was one of her fingerprints.

Murder at 15 Parson Street



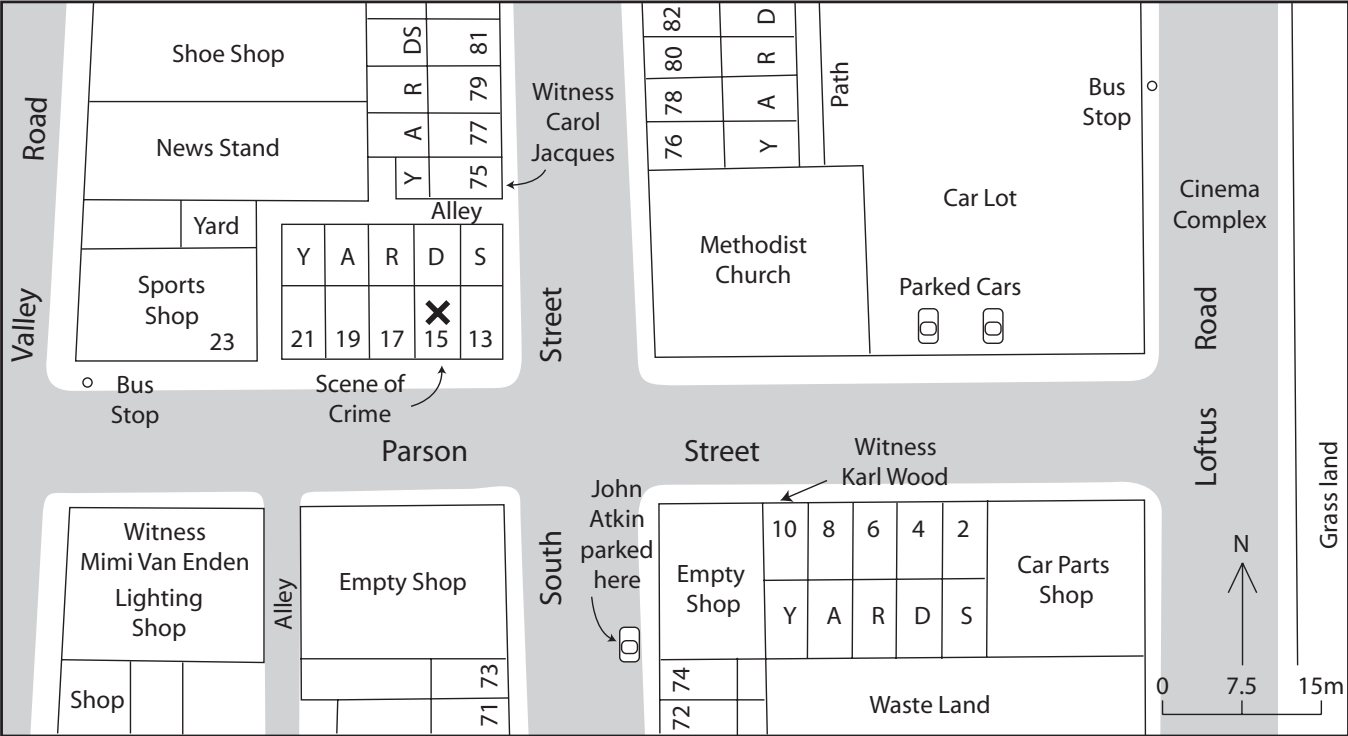
YOUR TASK – You are a detective in the homicide squad of the Harrington Criminal Investigation Branch, and are investigating a murder that has occurred in one of the offices of a vacant building in the city centre. You have questioned five suspects, and from the available evidence, you must determine the identity of the murderer and the probable motive.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task and the police interview statements, and make a list of the most likely suspects and their motives.
- 2 Make a list of the evidence that might help you to determine the identity of the murderer, and the method and motive for the crime. To whom does each piece of evidence point?
- 3 Based on the information you have accumulated, indicate who you think committed the murder, his/her motive for the crime, and explain how it was carried out.

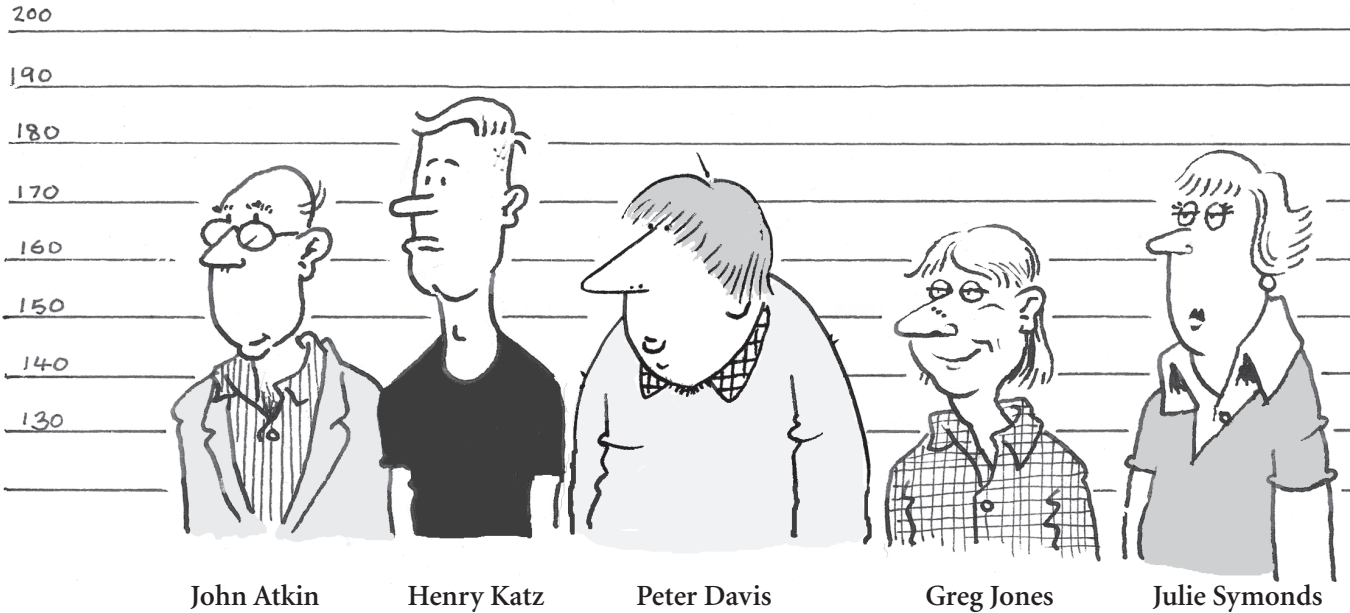
THE SCENE OF THE CRIME



Murder at 15 Parson Street



THE SUSPECTS



John Atkin

Henry Katz

Peter Davis

Greg Jones

Julie Symonds

POLICE REPORT

At 2.33 pm a telephone call was logged with 000 indicating that a gunshot had been heard coming from number 15 Parson Street. Three police vehicles were dispatched to the scene, the first arriving at approximately 2.45 pm.

On entering the building at 2.48 pm, Senior Constable Damian O’Hara and Constable Litea Mahina discovered the body of a deceased male, probably in his early 20s, lying on the floor of an office at the front of the building. He appeared to have been killed by a single gunshot to the chest.

It was apparent that the building, which had been vacant for about a month, was being used as a squat. Rubbish, empty take-away food containers and some personal belongings indicated that a number of people had been living there during the previous few days.

The area was immediately cordoned off, and a sweep made of adjoining buildings and grounds. Two males, Peter Davis and Greg Jones, were taken into custody, after being found hiding in a nearby car lot on Loftus Road. Davis and Jones admitted to having squatted at number 15, but claimed to know nothing about the murder. Jones had a deep cut across the bridge of his nose, and blood on his clothing. Jones and Davis stated that two other people, Julie Symonds and Henry Katz, had also been using the building as a squat. A warrant was issued for their arrest. However, at 4.35 pm both parties surrendered voluntarily to Harrington Police.

Police also arrested Mr John Atkin, who was discovered sitting in his car in South Street at 3.05 pm. Mr Atkin is the owner of several buildings in Parson Street, including number 15. On observing Mr Atkin in his car, the arresting officers approached him as a matter of routine, hoping that he might be able to help them with their investigations. Mr Atkin seemed extremely agitated, and upon asking him to step out of his car, the police officers noticed that he had what appeared to be blood spots on his suit, and that he also had an unlicensed gun in the glove compartment of his car.

Another gun was located in the bushes of the grounds of the nearby Methodist church. It had been recently fired and one chamber was empty. A single fingerprint was found on the weapon. A black tracksuit and beanie were found in a rubbish bin in the alleyway between South and Parson streets.

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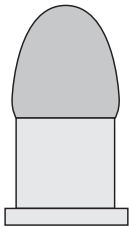
POLICE REPORT

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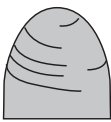
The deceased male was identified as Mr Grant Stanton, aka ‘Groover’.

A bank passbook was found among Mr Stanton’s belongings, which showed that he had deposited \$200 into his account on the day before his murder.

Two weapons were found near the scene of the shooting. One, an unregistered gun, belonged to Mr Atkin. The other, also unregistered, was discovered in some bushes in the grounds of the local Methodist church. Both pistols fired bullets of the same calibre. Forensic tests on the bullet removed from Mr Stanton produced the following results.



A bullet of the calibre fired by both guns.



A test bullet fired from the gun found in the church grounds.

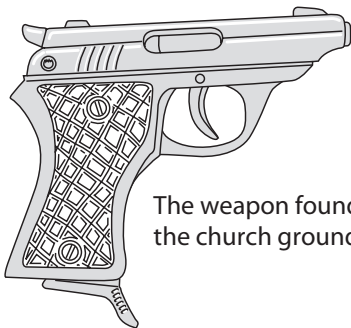
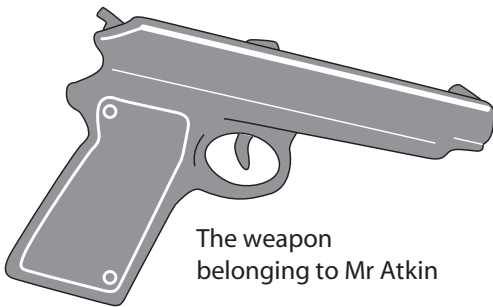


A test bullet fired from Mr Atkin's gun.



The bullet removed from Mr Stanton.

The two guns looked like this:



The weapon found in the church grounds

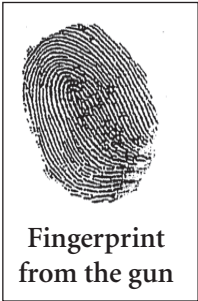
The blood on the clothing of both Mr Jones and Mr Atkin was established by forensics as having come from Mr Jones.

A spent cartridge case was later found in a public waste bin near the Bijou Cinema.

Two bullets were found at the crime scene. One in the body of the deceased, and one in the floor under an empty filing cabinet. Both came from the same gun.

There was also a pillow with a bullet hole through it.

The fingerprint on the gun was matched against the fingerprints of the five suspects and the deceased.



Peter Davis



Henry Katz



Julie Symonds



Grant Stanton



John Atkin



Greg Jones

Murder at 15 Parson Street



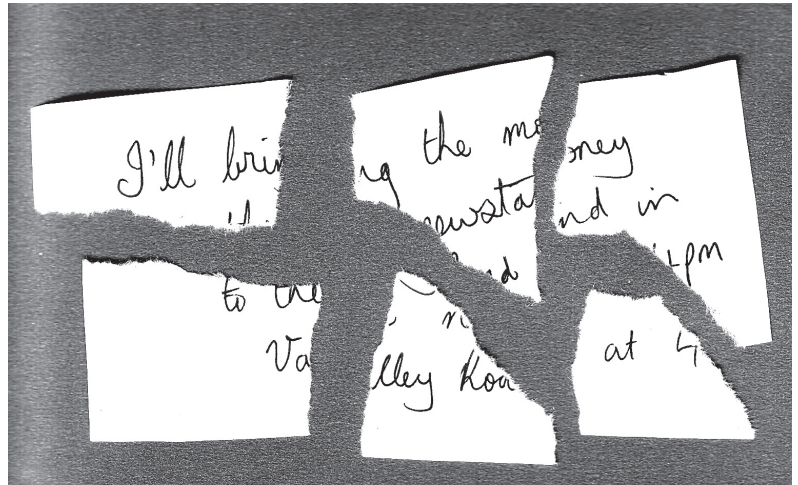
POLICE REPORT

(Continued from previous page)



The body was discovered at 2.48 pm. The forensic team examined the body at 3.30 pm. The normal body temperature is 36.9°C, but when dead the body cools at 0.83°C per hour for the first five hours, after which it cools at about one-fifth of this rate. At the time of examination, the body temperature was 33.6°C.

A torn-up blackmail note was found in the rubbish at the murder scene. Police matched it against samples of handwriting taken from the suspects, as well as a note found in the victim's belongings.



Peter Davis

It was a great
was up and the
You have never
it. The locals

Henry Katz

Meet 7pm at the
corner of Janice St
and Denise Road.
Try not to be late

Grant Stanton

Milk fried rice
white bread
butter ham
green beans
tomatoes lettuce

John Atkin

In his report on
that it is more widespread
originally thought. It
is spreading as we

Julie Symonds

I look forward to
seeing you on
Friday,

Greg Jones

I still haven't been
any work. At my age
any labouring and if
desk jobs they say

Murder at 15 Parson Street



SUSPECT STATEMENTS

Peter Davis, unemployed, of no fixed address

Greg and I are squatters. We travel from place to place looking for casual work. When we get sick of a place, we move on. It's not a bad life. We get to travel and see the sights, and all without any real responsibilities. If we don't like the boss or the work, we leave. The trouble is, it's hard to find a place to live. People don't trust us. Sometimes we stay in the backpacker hostels, but even that stretches our funds a bit thin, so whenever we can, we squat. We look around for any place that looks empty – a house, a factory or an office building – and we move in. A 'For Sale' or 'For Lease' sign is generally a good sign of a good squat. I suppose that technically we're breaking the law, but usually we're gone by the time anyone notices anything out of the ordinary, especially as we don't hang around during the day anyway.



Well, Greg and I have been staying at number 15 for the last four days. We get in and out through a broken window at the back of the building. When we arrived, three other squatters were already there. We only knew them as 'Katz', 'Groover', and 'Jules'. Katz and Jules seemed to be travelling together, but Groover was on his own. He was a nice bloke, and Jules seemed to like him. Jules acted like they were old friends. But Katz had a real attitude problem. He was always arguing and threatening. Just this morning Katz and Groover were at it hammer and tongs. Katz told Groover to keep his eyes off his girl, but Groover insisted that he wasn't interested in her. When we left to go job hunting, Katz was still in a foul mood. At lunchtime I met up with Greg. He had a really bad gash over his nose, but he wouldn't tell me how he got it. I've only known him for a few months, but I can't imagine him hurting anyone. It's just not in him, he's the placid type. Besides, as far as I know, he doesn't have a gun.

Now I'm normally a placid sort of guy too, but sometimes I lose my cool. I thought that Katz might have punched Greg, maybe because Greg had been nice to Jules or something. As I said, Katz seemed insanely jealous. You only had to look sideways at her when he was around and you could expect trouble. I suppose Greg thought that if he told me what had happened, I might have a go at Katz over it, and I don't reckon he wanted any trouble.

Greg Jones, unemployed of no fixed address

Pete and I have been mates for a couple of months. We move from squat to squat while we look for part-time work. When we arrived at this place, Katz, Jules and Groover had already moved in. Jules and Groover were okay, but Katz was an absolute freak. He seemed really jealous of Groover's good looks and easy-going manner. They'd had a few big arguments, but deep down I think that Groover was scared of Katz.

Anyway, this morning, after Pete and I had gone our separate ways looking for work, I bumped into Mr Atkin. He's the guy who owns number 15. He's tried several times to get us evicted, but I think that the law works too slowly for his liking. He went ballistic the moment he saw me, threatening to put his dogs on us if we didn't get out. I told him to push off, and he punched me on the nose. He said that if there was ever a next time, he wouldn't be so gentlemanly. Later, I met up with Pete for lunch. Although he wanted to know about my cut nose, I kept the details to myself. Pete does have a bit of a bad temper, and he could easily take matters into his own hands, and I didn't want any more trouble. Anyway, after we'd eaten, we started walking through a car lot towards the park at the end of Parson Street. It was then that we heard the gunshot. It really put the wind up us so we hid. Next thing we knew, we were being arrested by the police.



Murder at 15 Parson Street



Julie Symonds, unemployed of no fixed address

I met Katz soon after I ran away from home. At the time, it seemed like an adventure, but in reality it's actually turned out pretty badly. We have to scrounge for jobs, and sometimes we can't afford food.

Actually it was Katz who discovered this squat. He smashed a window at the back of the building, and got us in. The owner found out about us of course, but it's easier to get into these places than to be thrown out. He threatened us a couple of times, but as far as I know it's come to nothing.

Three other guys have been sharing the squat. Pete and Greg are okay, but they keep pretty much to themselves, as does Grant. He seems quite nice, but for some reason, Katz has taken against him, and they've had a couple of heated arguments. It took all my effort to keep them apart this morning.

At about 1 pm, Katz and I headed off to the movies. They were showing Tom Hanks and Audrey Tatou in the film of Dan Brown's book *The Da Vinci Code*. I really enjoyed it.

On our way back to our digs, we heard about the shooting, and Katz got really worried. He thought that because he and Grant didn't like each other, the police might think he'd done it. I reminded him that we had an alibi. They'd certainly remember us at the movie theatre, especially after I'd got into an argument with the usher for putting my feet on the back of the seat in front.



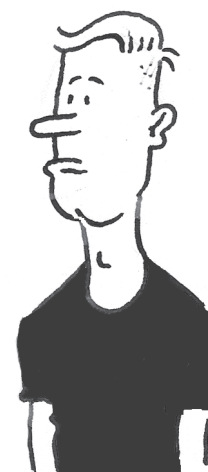
Henry Katz, unemployed of no fixed address

Jules and I go from squat to squat. It's a pretty easy going life, lots of friends and no responsibilities. Unfortunately, this time we got stuck with a real loser. Called himself Groover. We had a few arguments, mainly because he kept bothering Jules. She didn't like him, so I told him to back off. That's all there was to it.

On the day of the murder, I went out at about 10.15 am to get some food, and arranged to meet Jules at the local cinema at 1 pm. After, we found out that Groover had been done in. I reckon it was that property owner, Atkin. He'd threatened us a couple of times. He's got a real nasty temper!

As it turned out, it was lucky that we went to the pictures together, as Jules was with me for the whole afternoon. If she hadn't been able to provide me with an alibi, I think the cops would have tried to pin Groover's murder on me.

Actually, the film was a boring romantic comedy, and I ended up sleeping through most of it. I was out like a light almost the moment the lights went out, and I'd only just taken a few sips of the drink Jules bought me. She said she'd enjoyed the movie, but she seemed a bit uptight and distracted as we left the cinema. Her mood certainly didn't help my headache.



John Atkin, property owner

After I discovered that squatters were using my vacant building and turning it into a pigsty, I tried to have them evicted, but to no avail. So then I tried threatening them. Might have worked too, if it hadn't been for the one they called Katz. He put me in fear of my own life, so I borrowed a gun and started carrying it around for protection. I went to the building at about 11 am and heard a girl arguing loudly with someone. It had something to do with money. I didn't want to get involved so I left, as I was already late for an appointment with my solicitor. I admit I overreacted later on when I met up with one of the other squatters. Greg, I think his name is. He laughed when I told him I wanted them out, and I lost my temper and punched him in the face.



Murder at 15 Parson Street



WITNESS STATEMENTS

Peter Georgio, usher at the Bijou Cinema Complex

I remember them (pointing to the police photographs of Katz and Symonds). I thought they might be trouble right from the start. He arrived at the cinema about 12.45 pm, and seemed pretty irritated that she wasn't already there. Anyway, she came at about 1 pm.

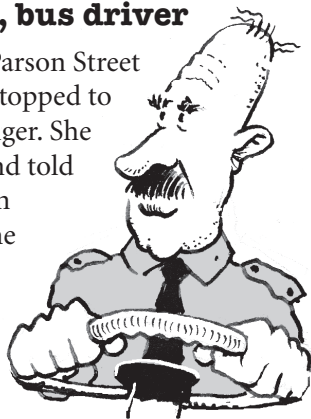
They'd only been in the theatre for about five minutes, when I had to tell the woman to take her feet off the back of the seat in front. She argued to begin with, but eventually did what I said when I threatened to call the police and have her removed. By this time her boyfriend was sound asleep, which was a relief, but how he slept through her shrieking I do not know.

During the middle of the film I noticed that she wasn't in her seat. I presume that she went to the restroom. I didn't see her return, as I'm responsible for keeping an eye on all four theatres. But I did see the pair of them leaving the cinema after the movie. She seemed quite agitated and irritable. When he asked her whether she'd enjoyed the film, she said yes, but she didn't seem to mean it.



Ahmet Massoud, bus driver

I had just turned into Parson Street from Valley Road and stopped to take on a female passenger. She seemed rather upset, and told me that a man had been murdered nearby. As she was getting off the bus at the Loftus Road stop, I could hear police sirens. It was all a bit scary.



Karl Wood, gift shop proprietor



I heard the gunshot at about 2.30 pm. I hurried to the front of my shop just in time to see a man run from the building diagonally across the intersection.

He was tall, well built, and wearing dark coloured clothing and a cap of some sort. Just as he got to the corner, I heard a scream. I'm only guessing, but I think it came from the same building as he had. Anyway, he hesitated for a moment, as if unsure of which way to go, and then turned and bolted up South Street. I dialled 000, and within 10 minutes there were police everywhere.

Mimi van Enden, lighting shop salesperson

I'd just finished serving a customer and was glancing through the shop window at the traffic going down Parson Street when I heard the sound of a gunshot. I knew it was a gunshot because my husband belongs to a pistol club and I often go to his competition events. Anyway, within a few seconds, a man ran from the front of number 15 and raced towards the intersection, turning left into South Street. I heard someone scream, but I didn't see who it was. Actually, I wasn't surprised that he'd come from number 15 because it's been vacant for a while, and I think it's being used by squatters. I'm not sure what time this happened, as I don't wear a watch, but it was probably around mid-afternoon. The man who ran from the building was tallish, and dressed in dark, loosely fitting clothes.



Carole Jacques, dress shop proprietor

It was about 2.30 pm when I heard the gunshot. Actually, it sounded more like a car backfiring. Then I heard someone scream, so I rushed to the front door, just as a person ran into the alley next to my shop. He was reasonably tall, about 180 cm or so, and was wearing a dark loosely fitting tracksuit, a black beanie pulled well down over his head, and a pair of wrap-around sunglasses.



Murder at 15 Parson Street – report

Write on the back of this page if you need more space.



An initial list of suspects and their possible motives (if any) for the crime

Name of suspect	Category	Comments

The name of the murderer is _____

Evidence on which your decision is based

Method and motive

[illegible]



8 Industrial espionage

TEACHER'S NOTES

In this mystery, despite the most stringent security precautions, the Luxor Motor Company's plans for the latest model of its number one selling sports sedan, the Luxor Lambretta, have been stolen.

The car is revolutionary in design, and if these plans fall into the hands of a rival, the company could lose its anticipated marketing advantage worth billions of dollars. Given the complex and seemingly foolproof security measures implemented by the R&D section, the suspicion is that it's an inside job.

The questions

- Who did it?
- How was it done?

The student information material includes:

- information about employees working for the company, including their various roles
- the transcript of the police interview with Ms Donna O'Brien, which throws significant light on the case in respect to the security system.

Note that understanding how the security system was circumvented will require some mathematical knowledge.

DEBATE TOPIC

The modern car is already a dinosaur.

Background

Cars have steadily and significantly changed people's way of life. Initially, they were used by the wealthy as a convenient form of transport. However, the mass production techniques introduced by Henry Ford eventually made cars the principal means of mass transport.

As the number of motor vehicles steadily increased, so did the associated problems. The increasing use of finite and irreplaceable resources, pollution, the need for elaborate and expensive road systems servicing large sprawling cities, and an unacceptable road toll have been just a few of these problems.

Have we gone too far to turn back? Is the modern motorcar an integral and indispensable part of life in a modern society? Are there ways and means of minimising or even eliminating these problems without 'throwing out the baby with the bath water'?

RELATED ACTIVITIES

Data security systems

You could hold a class discussion about the various methods of making data secure, and how these methods have changed over the past fifty years. You could have the students:

- list all the data security systems they can think of
- research how these systems operate, and the types of 'attack' they are designed to withstand
- list the pros and cons for each system.

Some discussion starters are listed below.

- In the past, valuable documents were stored in a safe or strongbox, usually in a secure location, with lockable doors, guards and an alarm system. However, accessing and using the data was more difficult, time-consuming, and often inconvenient.
- Valuable documents now reside on computer hard drives, protected by passwords and encryption. The information can be easily accessed, but the hard drive might be a server connected to a network of other computers.
- For the industrial spy, it's a matter of knowing the appropriate access codes and procedures. How individuals get access to this information is the key issue.

Coding, encryption and espionage

Ask anyone what they think a code is and they will probably say that it's a secret way of sending messages. In fact, a code is just a convenient way of conveying information, by any means at all. For example, the English language is a code. Only those in possession of the right decoding skills can understand it.

Ciphers are almost always secret and are used when information is intended to be conveyed only to those authorised to receive it. Ciphers have traditionally been the stuff of secret agents and war-time communication, creating a feeling of mystery, excitement and adventure. Now we speak matter-of-factly about the need for security when communicating with others, and use such procedures as encryption every day over the Internet and think nothing of it.

8 Industrial espionage

TEACHER'S NOTES

Research ideas:

- The history of codes and ciphers in military, political and economic espionage, in both fiction and real life
- How traditional types of ciphers work, and the techniques code-breakers use to decipher them. The role of the computer in this field could also be studied and applied.
- The different ways in which codes and ciphers are transmitted in order to keep the decoding methodology falling into the hands of others.

Students could also devise simple codes, and apply their skills to deciphering other codes.

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

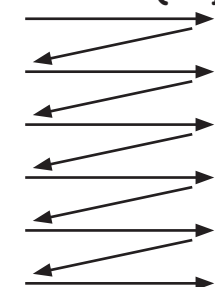
Who was responsible for the theft (10 points)

Anthony Baker (IT), his wife Christine Baker-Glasson (R&D), and her twin brother Craig Glasson (Security).

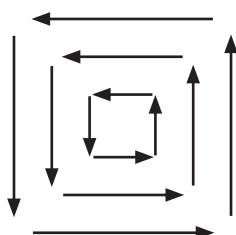
Reasons for your decision (20 points)

Carrying out such an audacious and difficult crime required the cooperative involvement of company employees who could discover the necessary combination number and passwords, and who could establish an appropriate timeframe for the crime, and then manipulate the building's security system to avoid detection. This meant at least three people had to work together – one from R&D, one from IT, and one from Security. The person from R&D needed to be a Team Leader in order to have access to the regularly changing passwords. Logic points to the three listed above as the most likely.

Likely method by which the crime was committed (20 points)



Prime numbers
at start of cycle



The numbers were then
cycled in three groups,
one number at a time

Craig Glasson used the security video surveillance system to watch the R&D Manager enter the safe's combination number each day. It didn't take long for him to realise that it was the current day's date in reverse order.

Each day the numerical passwords on the DVDs were also changed by the R&D Manager.

Christine Baker-Glasson kept a record of these passwords, noticing that the original password for her team's DVD was a series of five prime numbers.

Her husband, Anthony Baker, realised that at its starting point, the grid of passwords might have comprised a 'series' of consecutive prime numbers, and that these were then rotated each day in an anticlockwise direction as three separate groups.

This meant that the grid would not return to its original form for 4620 days (or more than 12.5 years). However, using this knowledge, Anthony Baker could determine the passwords for each DVD for any given day.

At a time when Craig Glasson was on evening duty, and the R&D staff had finished for the day, Christine Baker-Glasson entered her Manager's office, opened the safe and removed the DVDs.

Her husband, using the security system access information provided by Craig Glasson, successively disabled the computer-controlled security system to each area of the building long enough for his wife to finally exit without the DVDs being detected.

Later, using the password information and the stolen encryption disk, Anthony Baker was able to download the data from the individual DVDs to a single contiguous readable file. This was then re-encrypted using a different encryption system, and copied to tape, which could then be sold to the highest bidder.

Unfortunately, by the time that the theft was discovered, the disks were long gone. Donna O'Brien made the mistake of introducing her password system at the beginning of the cycle, rather than at some other point when the original relationship between the numbers would not have been decipherable.

YOUR TASK – You have been employed by the Calthorpe Private Detective Agency to investigate a theft from the Luxor Motor Company. The plans and specifications for the latest model of its top selling sports sedan, the Luxor Lambretta have been stolen, despite the most stringent security precautions. The design of the vehicle is revolutionary, taking a range of automatic operational functions to a completely new level, and setting the performance standards for at least the next decade. Should the plans fall into the hands of an unscrupulous rival company, the marketing advantage of a billion dollars of research and development (R&D) may be lost. Your job is to find out who stole the plans, how the crime was carried out, and whether the plans are still in the car plant.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

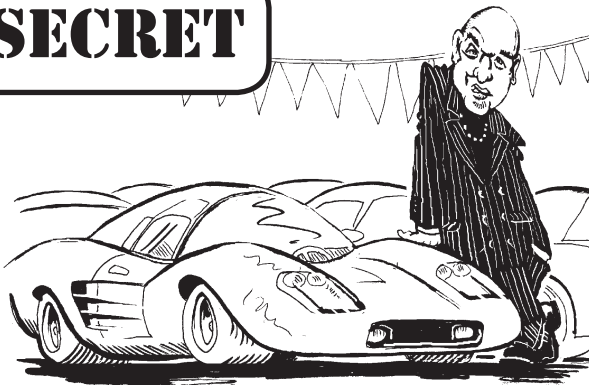
- 1 Read your task, the notes on the structure and operation of the Luxor Motor Company, and Donna O’Brien’s interview statement. Think about how you would go about committing such a crime, given the tight security measures involved.
- 2 Examine the employee list. Look for any connections that might point to the identity of the individuals involved.
- 3 See if you can figure out the methods by which the safe’s combination number and the DVD passwords were derived.
- 4 State who you think committed the crime, what their motives were, and how the crime was carried out.

The Luxor Lambretta

Features

- ☆ Petrol/electric motor
- ☆ Anti-lock breaking
- ☆ Independent self-parking
- ☆ Thief-proof security system
- ☆ Climate-control air conditioning
- ☆ Flexible shatterproof glass
- ☆ Power-assisted, computer-controlled GPS steering and safety system
- ☆ Cruise control and automatic speed regulation
- ☆ Computer-controlled four-wheel drive and independent suspension
- ☆ Dent resistant, colour-impregnated body panels

TOP SECRET



LUXOR AUSTRALIA – RESEARCH AND DEVELOPMENT



LUXOR AUSTRALIA EMPLOYEE DETAILS

Employee name	Date of birth	Gender	Dept	Address
Danny D'Angelo	5/5/71	M	R&D	5 Casey Drive Carrington 3099
Graham Arthur	14/4/55	M	R&D	Flat 4/181 Sheffield Cres. Gladstone 3189
Mary Azzopardi	12/9/71	F	IT	14 Smith Street Carrington West 3098
Anthony Baker	23/8/67	M	IT	231 Fletcher Drive Carrington 3099
Christine Baker-Glasson	13/5/68	F	R&D	231 Fletcher Drive Carrington 3099
George Boulis	29/1/75	M	M&C	75 Blake Street Gladstone 3189
Colin Brockly	11/7/72	M	Security	25 The Boulevard Gladesville 3180
Edwin Chng	24/2/62	M	R&D	18 Herbert Grove Oakton 3084
Carl Considine	18/11/59	M	R&D	137 Walter Court Gladstone 3189
Alan Edwards	31/5/75	M	R&D	47 Snell Cres Gladesville 3180
Colin Forbes	1/7/67	M	R&D	65 Poplar Avenue Oakton 3084
Craig Glasson	13/5/68	M	Security	Flat 2/16 Donnington Street Gladstone 3189
Guy Gregory	19/4/80	M	R&D	79 Doveton Road Gladstone 3189
Gerhard Holgart	7/8/69	M	M&C	16 Grantley Way Oakton 3084
Margaret Hughes	11/10/77	F	R&D	7 Porter Street Carrington 3099
Walter Hughes (Manager)	17/12/54	M	Security	15 Jacob Street Carrington West 3098
Helen Jedynak	13/3/73	F	R&D	88 Lincoln Cres Oakton 3084
Ray Jones (Manager)	5/12/74	M	IT	7 Worthington Street Gladstone 3189
Margaret Kelly	5/2/68	F	M&C	2 Shipley Road Gladesville 3180
Michael Kostos	23/1/73	M	R&D	21 Paddington Road Oakton 3084
Lisa Lan	17/7/72	F	R&D	6 Trentham Avenue Gladstone 3189
Brenda Lane	29/11/79	F	IT	17 Bishop Street Walthorpe 3187
Timothy Lee	10/4/68	M	Security	49 Winkley Way Oakton 3084
Jo Margaritas	18/11/73	M	R&D	121 Chifley Drive Carrington West 3098
Dan Minh	4/1/72	F	M&C	11A Badger Creek Road Gladesville 3180
Donna O'Brien (Manager)	21/6/72	F	R&D	6 Pantonville Cres Oakton 3084
Bekir Parmak	18/10/65	M	M&C	17 Curtain Avenue Carrington West 3098
Peter Raskovski	9/10/53	M	M&C	8 Pemberton Place Gladesville 3180
Jennifer Roberts	8/3/71	F	Security	36 Jackson Terrace Walthorpe 3187
Phillip Serkis	29/6/59	M	M&C	89 Kingsley Drive Walridge 3182
Elizabeth Sirianni	1/5/84	F	R&D	18 Paddington Road Oakton 3084
Casey Smith	5/9/79	M	IT	42 Collier Street Gladstone 3189
Jarrold Solway	31/10/73	M	IT	4 Denton Drive Oakton 3084
Terry Strachan	17/5/76	M	R&D	19 Drake Street Walthorpe 3187
Tony Sutcliffe	6/3/65	M	M&C	7 Briggs Street Carrington 3099
Michael Tandos	29/10/75	M	R&D	13 Blake Street Oakton 3084
Carlo Tanti (Manager)	30/11/61	M	M&C	21 Chipperfield Way Oakton 3084
Phillip Thomas	15/6/76	M	IT	17 Bishop Street Walthorpe 3187
Danny Thompson	12/8/66	M	Security	Flat 7/16 Donnington Street Gladstone 3189
Deng Tran	15/7/69	M	R&D	88 Wilton Road Carrington West 3098
Peter Whitelaw	27/4/80	M	IT	1 Crampton Cres Walridge 3182
Bevan Willard	21/8/71	M	R&D	22 Franklin Road Gladstone 3189
Frans Woldhuis	4/12/74	M	IT	14 Normanby Street Carrington 3099
Demir Yesmin	3/9/70	M	R&D	31 George Street Carrington West 3098



STRUCTURE AND OPERATION OF DIVISIONS OF THE LUXOR MOTOR COMPANY

Research and development (R&D)

The 20 people who work in the R&D section are divided into 10 teams of two. Each team works on a different aspect of the total project. Each has a leader who is responsible for setting the team’s daily work agenda as well as ensuring that the results of its efforts are saved to the IT server and the backup DVD. Only the R&D Manager (who also heads up one of the teams) and the Team Leader know the password necessary to access the team’s computer files. This password is regularly changed by the R&D Manager. No-one other than the R&D Manager has access to, or knowledge of, the overall product development plans. Members of R&D are forbidden to discuss any aspect of their work outside their team meetings. Failure to comply results in instant dismissal. By way of compensation, they receive an excellent salary package, and their jobs are highly sought after.

Information Technology (IT)

Nine people are employed in this division of the company. Their task is to maintain all computer equipment (desktop PCs, file servers, and the operation and integrity of the network and the email system), and ensure the daily backup of all relevant files. The supply of computer hardware and most software development is tendered out. All files are password protected and automatically encrypted, so that no company personnel (including IT staff) have meaningful access to sensitive material.

Security

The six people who work in this section maintain an overview of the company’s comprehensive video surveillance security system. They work in pairs, in three eight-hour shifts. Most of the security functions are fully automated. For example, movement throughout the building is governed by the use of a security card system that recognises appropriate access privileges.

EMPLOYEES OF THE DIVISIONS

Research & Development Laboratory

Note that team leaders are underlined.

<u>Graham Arthur</u>	<u>Christine Baker-Glasson</u>	Edwin Chng	Carl Considine
Danny D’Angelo	Alan Edwards	<u>Colin Forbes</u>	Guy Gregory
Margaret Hughes	<u>Helen Jedynak</u>	<u>Michael Kostos</u>	Lisa Lan
<u>Jo Margaritas</u>	<u>Donna O’Brien</u> (Manager)	Elizabeth Sirianni	<u>Terry Strachan</u>
Michael Tandos	<u>Deng Tran</u>	Bevan Willard	<u>Demir Yesmin</u>

Information Technology

Mary Azzopardi	Anthony Baker	Ray Jones (Manager)
Brenda Lane	Casey Smith	Jarrod Solway
Phillip Thomas	Peter Whitelaw	Frans Woldhuis

Security

Colin Brockly	Craig Glasson	Walter Hughes (Manager)
Timothy Lee	Jennifer Roberts	Danny Thompson

Maintenance and Cleaning

George Boulis	Gerhard Holgart	Margaret Kelly
Dan Minh	Bekir Parmak	Peter Raskovski
Phillip Serkis	Tony Sutcliffe	Carlo Tanti (Manager)



INTERVIEW WITH DONNA O'BRIEN, R&D MANAGER

D.O'B: This is the most ambitious project ever undertaken by the Luxor Motor Company. When in production, the Luxor Lambretti will be years ahead of its time, and far beyond anything on the drawing boards of our competitors. Imagine a vehicle that steers itself, knows exactly where it is at any given moment, regulates its own top speed according to the speed limit of the road on which it's being driven, can park itself, resists damage to its body panels at collision speeds up to 70 k/h, and never has to be polished, and rarely washed. Almost \$2 billion has been spent in developing a prototype, and now the plans and specifications have been stolen.

Investigator: Where were the plans kept?

D.O'B: The plans of the Lambretti are actually contained on eleven DVDs that provide descriptions, schema, and specifications of every single one of the 217 838 individual components of the motor vehicle. The DVDs are stored in a fireproof safe, with a six-digit combination code that only I know, and that I change on a daily basis.

Investigator: How are the plans on the DVDs upgraded as modifications occur to designs and theoretical operating specifications?

D.O'B: At the start of each day, I load the DVDs into a multi-disk unit attached to the R&D computer. It's a mainframe located in the IT division. Employees in each section of the division only have authorised access to the DVD that contains the plans and specifications relating to their area of operation. DVD1, for example, relates to bodywork, DVD2 to transmission, DVD3 to steering, and so on. DVD11 contains the master encryption code. Data from the DVDs is downloaded as required, and at the end of each day, modifications are uploaded to the master disks, which I remove and place in the safe. The working programs are then encrypted and backed up onto the mainframe. As I said, the encryption program is on DVD11, and without that disk, the backup files on the mainframe are unreadable.

Investigator: Is each DVD password protected?

D.O'B: Yes, and only the Team Leader and I know it, and just in case you intended to ask, I change the six digit password on a daily basis using a mathematical rotation method.

Investigator: Does anyone else have access to the programs containing the plans and specifications?

D.O'B: Technically yes, although it's a complicated process. Authorised members of the company's IT division have access to the R&D computer files, as it's their responsibility to keep our computer operational. However, as the encryption program is not on their mainframe, they cannot actually make any sense of the information these files contain.

Investigator: Now that the DVDs have been stolen, how can you access your encrypted files on the mainframe?

D.O'B: The company retains a second copy of DVD11. It's stored in a bank safety deposit box that can only be accessed by the company's CEO and me, acting together. The backup files are also protected. No R&D file can be deleted by anyone in the IT division without the IT Manager authorising the deletion. Nor can any R&D file be downloaded from the mainframe without the authorisation of the R&D Manager. So you can see that the system is securely protected.

Investigator: Except for the fact that the DVDs have been stolen.

D.O'B: Well that's also a problem for the thief. You see, each DVD contains a microchip that sounds an alarm at any point of entry or exit to the building. Every door and window that can be opened has been alarmed by the company's Security division.

Investigator: So you're saying that having stolen the DVDs, the thief would have been unable to remove them from the building?

D.O'B: That's right. However, our security people have completed an electronic sweep of the building, unfortunately with no success. I don't understand why, but it appears that the DVDs have just vanished.

Investigator: What was to stop the thief from copying the DVDs to another set of disks?

D.O'B: That's not possible. By an ingenious method, with which I'm not really familiar, the information contained on the DVD actually appears to exceed a DVD's normal maximum storage capacity, and since the data exists as a single file, you can't copy it.

(Continued over)



Investigator: Well, according to the computer system controlling the safe, it was opened four hours and 15 minutes ago. Let’s hope that the thief hasn’t had time to remove the disks from the building. By the way, what was the combination number of the safe on the day of the robbery?

D.O’B: It was 3 0 9 0 0 3. Do you want to know how I derive the new combination number each day?

Investigator: That won’t be necessary. I already know how you do it. What were the passwords on the first day you issued the DVDs to the Team Leaders?

D.O’B: Just give me a moment or two and I’ll work it out for you.

DVD1	2 3 5 7 1 1
DVD2	1 3 1 7 1 9
DVD3	2 3 2 9 3 1
DVD4	3 7 4 1 4 3
DVD5	4 7 5 3 5 9
DVD6	6 1 6 7 7 1
DVD7	7 3 7 9 8 3
DVD8	8 9 9 7 1 0
DVD9	1 1 0 3 1 0
DVD10	7 1 0 9 1 1
DVD11	3 1 2 7 1 3

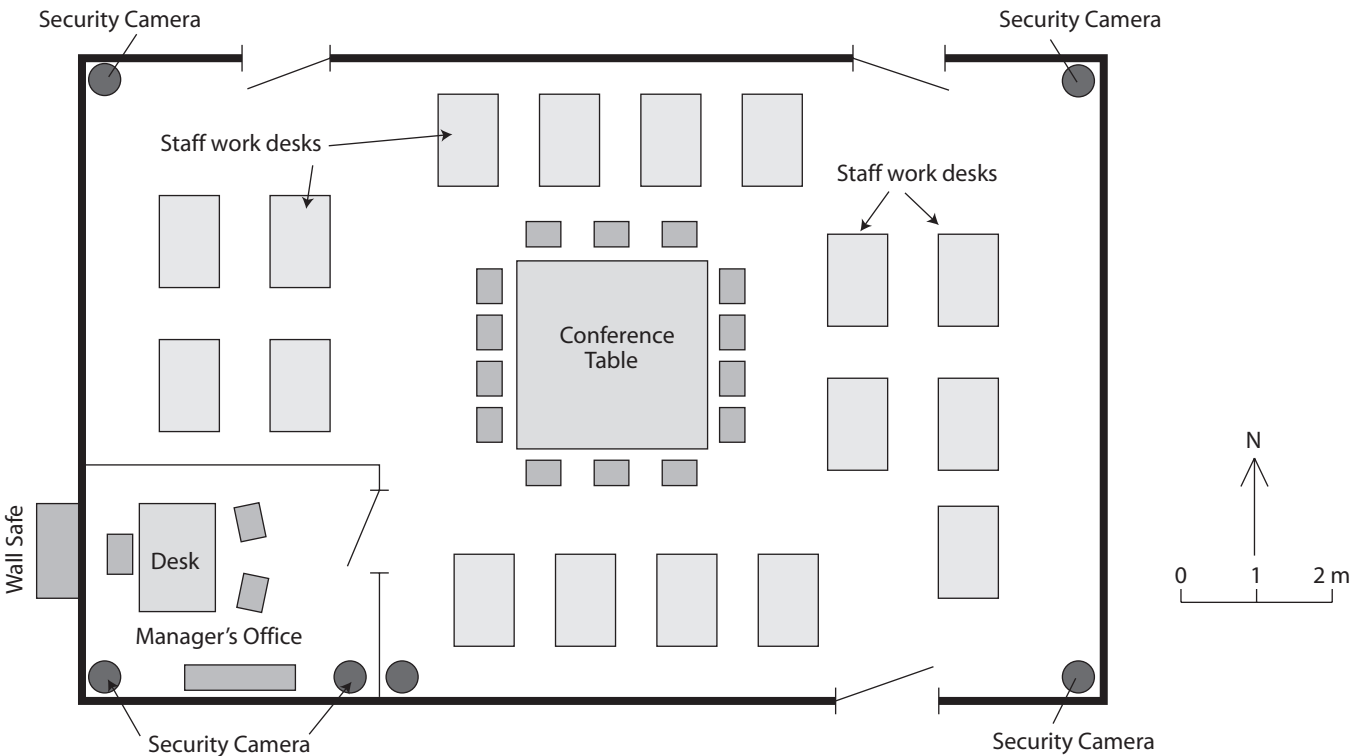
Investigator: And what were they on the day after?

DVD1	3 5 7 1 1 9
DVD2	2 3 1 7 1 1
DVD3	1 3 9 1 3 3
DVD4	2 7 2 3 4 9
DVD5	3 7 4 7 5 1
DVD6	4 1 5 9 7 3
DVD7	6 3 6 7 8 0
DVD8	7 9 7 3 1 0
DVD9	8 1 9 0 1 1
DVD10	1 1 0 9 1 3
DVD11	7 3 1 2 7 1

Investigator: Unfortunately, I can see exactly how you determined the 11 different passwords and how they varied from day to day.

D.O’B: Can you? But even so, you’re seeing the whole grid. Each team leader only saw their own individual DVD password, and there were 4620 different combinations.

Investigator: That’s true, but an astute person might have been able to work out the overall method from just a few examples, and you did begin at the start of the cycle. That was your mistake.



Industrial espionage –report

Write on the back of this page if you need more space.



Who was responsible for the theft? _____

Most likely motive for the crime

Reasons for your decision

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

Likely method by which the crime was committed

(Indicate whether or not you think the plans are still in the car plant)

[illegible]



The haunted house

TEACHER'S NOTES

In this mystery, a number of unexplained phenomena have occurred at an old, derelict mansion on the outskirts of Pentonville.

The questions

- Who or what is behind these events?
- If a person is responsible, what is that person's motive?

This mystery differs from earlier ones because:

- there are several quite feasible explanations to the mystery
- the final question requires a creative response from students.

Who or what is behind these events and why?

The student information material includes:

- relevant articles from the local newspaper, *The Daily Bugle*
- views from ordinary citizens through the paper's Letters to the Editor column.

This haunted house mystery has its basis in superstition, misreporting by the local newspaper, self-serving gossip and creative illusion. The author does not hold the view that the house is actually haunted. It is possible, but the author knows of no scientifically confirmed case of supernatural haunting.

RELATED ACTIVITIES

Research – hauntings

- Students could consider how one could scientifically test a 'haunted house' claim.
- Students could consider ways in which the illusion of a haunted house could be created.
- Students could research famous cases of ghosts and hauntings, and try to work out ways in which these events could have been faked.

Creative writing

Most students enjoy stories about the supernatural, and probably also enjoy writing their own. These topics should stretch the imagination of even the brightest student.

- 1 Gravestone Mansions has been haunted by a wailing, chain-clanking, 'head-under-the arm' ghost for the last 500 years, and the future

looks equally as bleak. Unless it can find someone prepared to voluntarily take its place, the ghost faces another 500 years of haunting.

- 2 Two hundred years into the future the world faces a terrible threat to its survival as a result of a seemingly unimportant event that occurred in the year 2006. Harrison Jones has discovered a way of projecting his image into the past in the hope of changing the course of history. However, his diaphanous appearance at the other end of the time portal encourages the owners of 124 Dilbert Place to imagine that their home is haunted. Time is running out ...
- 3 James Bent knows that there's a ghost living in his bedroom wardrobe, but nobody believes him. The trouble is, it only shows itself at the witching hour, and he's the only one who can see it. It seems so unhappy. What does it want?
- 4 Ghosts haunt the living, but what haunts the ghosts? And what if IT discovers the world of humans? (This one is for students who would like to write their own Stephen King thriller).

DEBATE TOPIC

The claim that ghosts exist has no substance.

Background

There have been many famous ghost stories throughout the ages, even supported in some cases by reasonably credible eye witness accounts as well as the odd photograph or two. To date, none has been scientifically verified, despite considerable time, effort and funding. By the same token, there have also been some very intriguing cases that seem to defy rational explanation.

RESOURCES

Here are some websites you might like to check:

- <http://theshadowlands.net/famous/>
(Some of history's more famous ghosts)
- www.hauntedhome.com/theHouse/Lore/goblins.htm



The haunted house

TEACHER'S NOTES

SAMPLE REPORT

Note: The marking scheme is a suggestion only.
You may not want to score the work at all.

Who was most likely responsible for creating the impression that Lansgrove Manor was haunted? (5 points)

Ms Angela Palmer

Why did the perpetrator carry out this deception? (20 points)

- 1 Angela Palmer is an entrepreneurial property developer who stands to make a lot of money if Lansgrove Manor is sold to a supermarket chain that already owns a large parcel of adjacent land.
- 2 Lansgrove Manor prevents the completion of the rail link between Pentonville and Pentonville East Railway Stations.
- 3 Angela Palmer hopes to buy the land cheaply because of the ghost story, and then resell it at an inflated price to the supermarket chain. She uses speakers and external light sources to create the ghostly apparitions, and writes to the local newspaper as Mrs Mary McMahon in order to stir up gossip and even fears about Lansgrove Manor being haunted. Her aim is to discourage potential buyers. The newspaper is quick to give its own slant to the 'so-called' facts, such as the distorted views attributed to the Gravenhursts and the police.
- 4 There is, of course, the additional possibility that Frederick Parson is an accomplice. This would mean that the supposedly inexplicable ghostly apparition referred to in his report was created by him from within the room, not externally as suggested in the next section. However, his being an accomplice seems to contradict the fact his report exposes the various other tricks used to give the impression that the house is haunted.

How was the ghostly apparition that Frederick Parsons was unable to explain created? (25 points)

The ghostly apparition was most likely caused by an externally projected image on the blind covering the window of the room that Frederick Parsons was occupying at Lansgrove Manor. The blind was made from material that did not allow the passage of light, so that what was projected from the outside did not show through the blind to the inside. Therefore, Frederick Parsons would not have seen the apparition.

On the other hand, if Frederick Parsons was Ms Angela Palmer's accomplice, then the image could have been projected from within the room by Parsons himself. In this case, it would have been desirable for the blind to be translucent, thereby giving the impression of ghostly apparitions to any passer-by.

Suggest other ways in which such an illusion might be created. (10 points)

Hopefully, this open-ended question will produce a number of different suggestions for methods of creating ghostly apparitions. Some students have suggested a liquid crystal flat screen carefully mounted in the window, lighting of the sort used in Christmas decorations, and so on. The class should carefully examine the plausibility of each suggestion. Some will be impractical, others impossible.

The haunted house

9



YOUR TASK – You are an employee of the Ace Paranormal Enquiry Agency and have been assigned to investigate a series of unexplained phenomena at an old, derelict mansion on the outskirts of the city. The owner is keen to sell the property, but the reputation of the house has discouraged most potential buyers. The only offer to date is well below its market value, especially given its strategic location. Your task is to determine whether or not the house is haunted, and if not, who is behind the deception, what the perpetrator's motive might be, and how the ghostly apparitions are created.

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task.
- 2 Read the extracts from *The Daily Bugle*, which contain information that will help you to reach a solution.
- 3 Read Frederick Parsons' report.
- 4 Look at the map of Fairview Estate. This indicates the strategic position of Lansgrove Manor.
- 5 From this information you should be able to confidently speculate on whether or not the hauntings are a hoax, and if so, who is responsible.



The Ace Paranormal Enquiry Agency

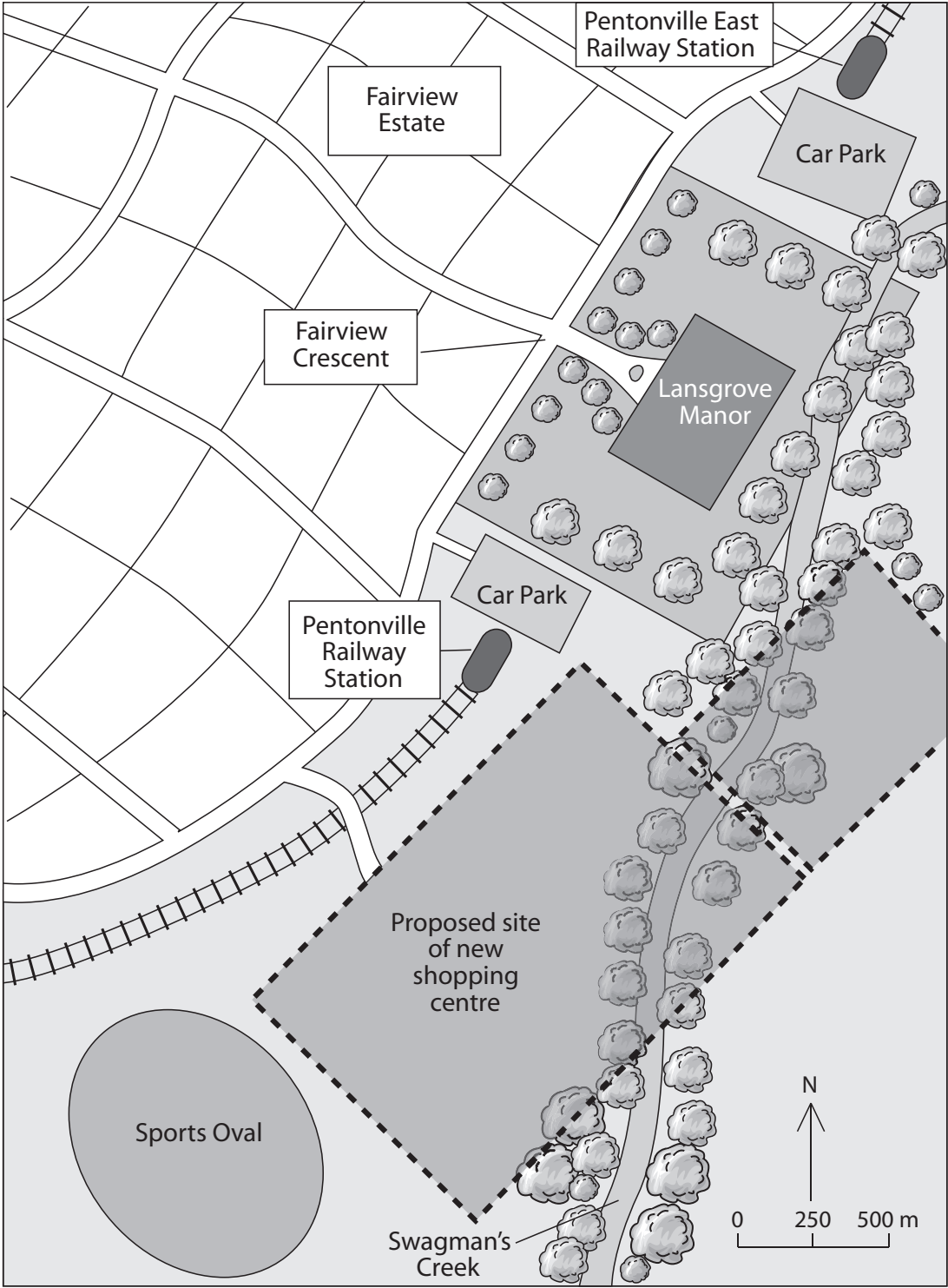
Established 1998



Lansgrove Manor



MAP OF FAIRVIEW ESTATE





The Daily Bugle

Thursday 4 August

The House on Haunted Hill

by Kylie Roberina

Is the house at 13 Fairview Crescent Pentonville really haunted?

That is a question which will hopefully be resolved by the Ace Paranormal Enquiry Agency when its top investigator takes on a case that has mystified local authorities for many years.

Since the dead body of the previous owner, Mr Clayton Lansgrove, was discovered in one of the upper rooms of the house seven years ago, there have been many reports of unearthly screams and moans emanating from the now derelict mansion.

A nearby neighbour, Ms Clarice Standish, claims to have seen ghostly apparitions through the half-shuttered windows of some of the upper rooms. Another local resident, Mr Basil Gross-man, believes that such ridiculous rumours have seriously eroded Pentonville property values.

The present owner, Miss Cecilia Lansgrove, granddaughter of the deceased, lives in England and is keen to dispose of the Lansgrove estate. However, there has been little interest, especially at its current \$3.5 million price tag.

'Lansgrove Manor', as it has been popularly known since its construction in the early 1800s, has not even been able to keep a tenant for longer than a week.

Mr Tim & Mrs Sally Gravenhurst, formerly of Lockington, were sceptical of the house's reputation and thought it might be a 'hoot' to live there, given the rent was so low. 'We only lasted four days', Joe said. 'It was terrible at night. Moaning sounds as though from a person in pain would come from the upper rooms. Despite our attempts to heat the place, there were parts of the building that were as cold as a grave.'

Ms Angela Palmer, wealthy entrepreneur and property developer, believes that no one in their right mind is likely to buy a haunted house. She says that she is personally willing to pay \$500 000 for the old house and then tear it down. 'I'd consider it a service to the local community,' she said. 'Perhaps I'd turn it into a children's playground.'

A representative of the Ace Paranormal Enquiry Agency stated that they just dealt in the facts, and used a range of sensitive scientific equipment such as special infra-red cameras and subsonic microphones to analyse so-called paranormal events. 'We are yet to prove the existence of ghosts', he concluded.



Above: An artist's view of Lansgrove Manor

THE FACTS

Clayton Lansgrove, a widower, was an eccentric millionaire who lived the life of a recluse in his mansion at the end of Fairview Crescent. His only son ran away from home when quite young, and Clayton had not seen him for almost 40 years. We have learned that the son, Percival, made his fortune in England, where he married, and had a daughter.

Seven years ago, Clayton Lansgrove was found deceased in an upper room of his house. He was only discovered after neighbours became suspicious at the constant howling of his dog, an Irish wolf hound. It was said that he had died of a heart attack, brought on by acute fear, the cause of which was never determined. Police said that the look on the deceased's face was one of absolute terror. The house itself was in a state of considerable disrepair, and uncollected rubbish was strewn about the various rooms.

It took many months to trace the next of kin, Miss Cecilia Lansgrove, whose own father, Mr Percival Lansgrove, had been killed in an industrial accident a year earlier.



The Daily Bugle

Friday 5 August

TV AND RADIO GUIDE – PAGE 49

Public Notices

Notice is hereby given that a major shopping centre is proposed for construction on the commercial site located at the south end of the Fairview Estate, Pentonville. This centre will include major hardware, furniture and whitegoods retailers, as well as several supermarket and fast food outlets. Future expansion will include a multi-storey hotel/motel with undercover parking facilities, a cinema multiplex, a leisure centre and children's playground, as well as an extension to the rail line to link Pentonville and Pentonville East railway stations. Details of this proposal including all plans and specifications are available for perusal at the offices of the Glendale City Council.

Additional information may be obtained from the offices of the Palmer Construction Company, 146-150 Hampton Street Glendale (Tel: 7767 1999).

Important – Anyone who has knowledge of, or has personally experienced, paranormal phenomena at, or adjacent to, the site located at Fairview Crescent Pentonville (also known as 'Lansgrove Manor') should contact the Ace Paranormal Enquiry Agency. A financial remuneration will be provided for any information that proves useful to the company in its investigations.

Telephone us now on 7767 2581.

Letters to the Editor

The facts

The real reason that we left 'Lansgrove Manor' was because the house was too cold and draughty, and the building never stopped creaking. At night we couldn't get to sleep for the noise.

Tim & Sally Gravenhurst (Lockington)

I take strong exception to the report in yesterday's *The Daily Bugle* suggesting that police had observed a look of terror on the face of the deceased, Mr Clayton Lansgrove, the former owner of 'Lansgrove Manor'. In fact, as I reported to the Coroner as the attending officer at the time, Mr Lansgrove looked seriously emaciated, which gave his face a skeletal appearance. I found nothing to suggest that he had died of anything other than natural causes.

*Inspector James Williams
(Glendale Police)*

It's all a fake!

All this talk about haunted houses is just a beat-up designed to sell newspapers. What about some real news?

Mary Jenkins (Lockington)

Good timing

Is it merely a remarkable coincidence or a sinister plot? Who wants to live next to a haunted house? No one! So the adjacent land goes cheaply. Then, surprise, surprise, the council rezones the land for commercial use, and suddenly we have a proposed shopping centre. I smell a rat.

Barry Bagsworth (Pentonville)

I told you so!

I've said repeatedly that Lansgrove Manor is haunted. The death of old Clayton Lansgrove was directly caused by the evil spirits that have inhabited the house for more than 200 years. How many other tenants have to die before the authorities do something about it? Act now!

Mary McMahon (Crayhurst)

The Medical Report

As the doctor who certified the death of Mr Clayton Lansgrove, I can assure the people of Pentonville, as I assured the Coroner, that Mr Lansgrove died from natural causes as a result of old age and increasing ill-health. His haggard appearance was accentuated by a state of malnutrition probably brought on by his inability to adequately care for himself. I observed no evidence of a stress-induced heart attack.

Dr Grant Wilson (Lockington)

Does she really exist?

I have searched the electoral roll as well as the telephone directory for the name of Mary McMahon, a frequent writer to this column, and one who seems intent on promoting the ridiculous notion that Lansgrove Manor is haunted. I'm convinced she doesn't exist. What then is the real motive of this mystery writer?

John Harvey (Pentonville)

The Daily Bugle stands by its version of these events. (Ed.)

The Ace Paranormal Enquiry Agency Established 1998



Report on claimed paranormal occurrences at 13 Fairview Crescent, Pentonville (also known as 'Lansgrove Manor')

Five days (and nights) were spent examining the target building and testing the claims made by respondents to the public notice appearing in *The Daily Bugle* on 5 August, including:

- 1 Unexplained coldness in some passageways and rooms, and particularly in the room in which the dead body of the owner, Mr Clayton Lansgrove, was discovered on the morning of 24 April.

The house is very old and draughty, and my equipment was not able to detect any abnormally cold areas that could not be explained by this fact. There was no part of the building that was not able to be heated to a constant and comfortable temperature using a portable gas burner. However, the house's own heating system, a recently installed ducted reverse cycle air conditioner, had a faulty thermostat that occasionally caused it to switch unexpectedly on and off, and from heating to cooling.

- 2 Ghostly apparitions observed at night through upper floor windows.

While some of these observations might be explained by the reflection of moonlight from windows, there was also evidence of human intervention. In several of the upper-floor rooms, spherical light reflectors similar to those found in nightclubs had been concealed within chandeliers in such a position that they were in line of sight through windows. A laser beam directed from a source external to the building could produce the types of ghostly apparitions described by respondents.

- 3 Ghostly sounds such as moans and screams, the agitated murmuring of a number of voices, continuous sobbing and the clanking of chains.

Many noises were caused by the house moving as timber and metal contracted during the cold nights. Some suggestions were clearly the result of overly active imaginations. However, again there was an indication of human intervention. Speaker cables were located in some of the air conditioning ducts, and although the speakers themselves were not in place, it was obvious that the screws used to attach the vent covers to the duct outlets had been tampered with on a number of occasions.

Conclusion

Some of the claims made by respondents can be explained by natural causes, and are generally those relating to the age and state of disrepair of the building. However, there was also strong circumstantial evidence to suggest that someone was trying to create the impression that the house was haunted. So-called paranormal incidents seemed to be most prevalent when potential buyers took up the estate agent's offer to spend a few days in the house. I recommend, therefore, that my findings be referred to the local police for further investigation.

However, there was one incident that occurred during my stay in the house, for which I have no explanation. Apparently while I was asleep in the room in which Mr Clayton Lansgrove died, a passing neighbour observed a ghostly figure walking slowly back and forth in front of the window. I had set up a motion detecting camera and microphone in the room before going to bed, but these devices recorded nothing unusual during the night. Besides, I had drawn the blind before retiring, so it would have been impossible for an outsider to see into the room. Normally, I would have put it all down to the imagination of the observer, but in this case I regard that person as a highly reliable witness.

Frederick Parsons

Investigator

ABN 123 456 789

72 Kings Road Glendale 3822

Ph: 7767 2581



The haunted house – report

Write on the back of this page if you need more space.



Who was most likely responsible for creating the impression that Lansgrove Manor was haunted?

Why did the perpetrator carry out this deception?

Frederick Parsons was unable to explain the ghostly apparition. How was it created?

Suggest other ways in which such an illusion might be created.



In this mystery, Jarrah must predict the weather in order to decide whether to hold his brother's 21st birthday party outdoors or indoors. His decision has important financial ramifications.

By accessing the Internet and his school library, Jarrah is able to come up with:

- a three-day weather forecast relating to the time of the proposed party
- a weekly weather forecast
- weather trends over previous decades.

But Jarrah knows that even the experts get it wrong from time to time, and so he also considers some beliefs that have guided people in the past when making predictions. Consequently, his research includes some personal observations as well as the results of some interviews with family and friends.

DEBATE TOPIC

Global warming is just another baseless 'doomsday prophecy' built on an exaggerated misrepresentation of the facts.

Background

One of the most controversial scientific issues at the moment is climate change, or more specifically, global warming. At the extremes of this debate we have those who claim that unless positive steps are taken to dramatically and quickly curb the current level of greenhouse gas emissions, a range of cataclysmic global effects will seriously threaten future human existence. Then there are those who hold the view presented in the debate topic.

It might seem that this is an argument that should be able to be resolved by an appeal to the facts; however, if you follow the debate it would seem that even the so-called facts are in dispute. There are those who hold the view that current changes in global climate are just examples of natural variation occurring within normal long-term cycles. They believe that the sun is the principal determinant of weather, and that in any case, technology and human ingenuity will find a solution without the need for a dramatic worldwide change of lifestyle.

Developing nations want an opportunity to enjoy the fruits of economic growth, despite potential damage to the environment. In the

cut-throat world of international trade, the developed countries understandably don't want to place themselves at an economic disadvantage by voluntarily reducing their green house gas emissions – thereby reducing their competitiveness through higher production and energy costs – unless their competitors do too. This is the so-called 'level playing field argument'.

However, the balance of scientific, political and community opinion is increasingly moving towards the view that we must deal with the problems presented by global warming and climate change before it is too late. The difficulty most of us have in coming to terms with the notion of global warming is that the effects are gradual and not clear cut.

RELATED ACTIVITIES

A simple example of trying to predict the weather is a means of introducing students to the science of meteorology, and the larger socio-economic issues relating to global warming and climate change.

Meteorology

What causes our weather? How do we measure it? How do we predict it? There is a worldwide billion-dollar industry employing a host of well trained scientists using the latest in technology. But, does all this science and technology deliver the accurate forecasts?

Students might consider the accuracy of their local weather forecasts over a period of a month in respect to temperature, rainfall, humidity, wind speed, and so on. Long-term forecasts are available on the Internet, and their reliability could be examined as well. Then there is the variation between current weather patterns and long-term trends. To what extent does the current weather vary from previous weather patterns for the same period? Investigating this will help students understand the complexity of the science of meteorology.

Beliefs and superstitions

Beliefs about how to forecast weather based on observations of animal behavior and the environment can be found in the Bible and even



in Shakespeare's plays. It would be interesting to consider why people like farmers and sailors, who had a vested interest in the outcomes, based their forecasts on what they saw in nature.

Why did people rely on what they saw around them and why were domestic and farm animals more likely indicators of weather? Did these observations link to what may naturally happen in particular seasons? Can aching bones and joints really be linked to changes in air pressure or the likelihood of precipitation? Students may like to devise an experiment to find out if there is any evidence of links between human health and the weather.

Mathematics – probability

This uncertainty of weather forecasting also lends itself to the mathematical study of probability (the likelihood of the occurrence of a particular event). What probability, for example, might we assign to each of the possible outcomes Jarrah faces? What is the likelihood of getting each of the different types of weather on the day of the party?

You might determine that there is a 40% chance of rain, and an 80% chance of strong winds. You could use the summarised conclusions of the students to determine such probabilities, or base them on established weather trends from similar periods in the past.

Resources

There are many references on weather to be found in most school libraries; however, one particularly valuable source of information is the Bureau of Meteorology's website www.bom.gov.au. From this site you can investigate virtually any aspect of the weather you can imagine (descriptive information, maps, forecasts, etc). One particularly useful site for students is 'Learning about the weather' (www.bom.gov.au/lam)

Other useful web sites include:

- global warming basics – www.nrdc.org/globalWarming/f101.asp
- alternative views – www.globalwarming.org
- weather forecasting and related issues – www.australianweathernews.com/forecast_forecasts_longrange.htm

- On long range weather forecasting – www.bom.gov.au/climate/ahead

SAMPLE REPORT

Note: The marking scheme is a suggestion only. You may not want to score the work at all.

What is your recommendation to Jarrah? Should the party planned for Saturday afternoon be held indoors or outdoors? (10 points)

Jarrah should hold the function outdoors.

On what do you base this recommendation? (40 points)

While it is true that October is a month in which the weather can be changeable, and there are quite good rainfalls at this time of year, the weather forecasts for the short-to-medium term suggest that the Saturday afternoon of the party should be fine.

There may be some rain early on, but it is expected to clear to a fine day by late morning. Light winds and a top temperature of 23°C suggest that the conditions outside will be very comfortable.

The forecast cold and wet spell is not expected to arrive before Sunday, so the odds on fine weather for the party are definitely in Jarrah's favour. Long-term trends indicate the probability of rain; but on the other hand, the anticipated effects of La Niña contradict this.

Temperatures have certainly been above average in recent years, so this is promising. Even if there is a modicum of truth in the superstitions Johnny has researched, they appear to be contradictory and inconclusive.

The personal suggestions of Jarrah's friends and relatives also do not carry much weight as they are based more on feelings and impressions than facts and figures. Even Dheran himself supports the idea of holding the function outdoors.

Unfortunately, should it rain, Dheran's suggestion of the rotunda may not work, depending on the number of guests invited to the function. However, having a back-up plan is always a good idea.

YOUR TASK – Since his Mum has been unwell, the job of organising his older brother Dheran's twenty-first birthday party has fallen to Jarrah. It is a big responsibility, and one that Jarrah really wants to get right. His family is not wealthy, and while Dheran said that he didn't want anything special or too expensive, Jarrah knows that his mum wants her sons' twenty-first birthdays to be occasions to remember. She has been putting aside a small amount of money each week over the years to make this possible. Dheran's share only added up to five hundred dollars, but Jarrah feels that, with careful planning, he can put on a really good party for this amount. The trouble is – the weather.

Hiring an indoor venue would put a big dent in the birthday fund. Having the party during the afternoon at the outdoor function area at a local park would mean that Jarrah could spend more money on food and entertainment. He'd heard that weather in October was often unpredictable, and he knew that he'd have to decide about the venue well before the event. What should he do?

This activity tests your ability to read with understanding, to think logically, and to express yourself clearly and concisely. Merely knowing the correct answer to the main question is not enough. You also need to explain how you arrived at your conclusion, and on what evidence you based your decision.

Follow these steps

- 1 Read your task.
- 2 Read Jarrah's diary entry.
- 3 Read all the weather prediction information.
- 4 From this information, decide whether the party should be held outdoors or indoors, and explain your decision.

Jarrah's diary entry

Dear diary,

Here I am, planning a 21st party – for Dheran, of course. And to do it properly, I have to know what the weather will be like on Saturday. I don't mind trying to guess, but there must be information out there that will help, and increase my chance of guessing right! There are lots of theories about predicting the weather, but are they accurate? Are they based on facts?

They say in this city that you can get four seasons in a single day, and that you know when it's summer because the rain is warm. But I'm not so sure about that now. There has been a lot of talk lately about global warming and climate change, and I've heard that the weather is becoming more unpredictable.

If I rely on an outdoor venue and it rains, it'll be a disaster – Dheran's party will be completely ruined. But hiring an indoor venue will cost a lot of money. What should I do?

THE RESULTS OF JARRAH'S INTERVIEWS



UNCLE HADIB October's a pretty unpredictable month. When I was a boy we seemed to get a lot of rain in October. I'd have the party indoors.

GRANDPA I reckon it'll be raining by Saturday, my bunions are starting to ache and that's a sure sign that the weather's going to change.

JARRAH'S MATE PAULO We've been studying global warming, and the scientists reckon that average temperatures are going up and rainfall is declining. I think it's a safe bet for a dry Saturday.

AUNTY MARY When the birds go quiet, bad weather is on the way. I agree with your Uncle, have the party indoors.

DHERAN Look, the odds suggest a fine afternoon, so my suggestion is to confirm the booking for the outdoor recreation area. If it does rain, which I don't think it will, we could use the rotunda.

Jarrah's observations of events during the last week before the party, and their relevance to his researched list of superstitions.

- 1 He noticed the cat licking her fur against the grain on at least two occasions, and on at least one of these occasions it was sitting on the table looking out of the window. He was not certain whether it sneezed at this time, or was coughing up a fur ball, as he was too far away to be sure, but it certainly gave a small convulsion. His research about cat superstitions suggested that taken separately, each of these meant rain. All together might mean the onset of a storm.

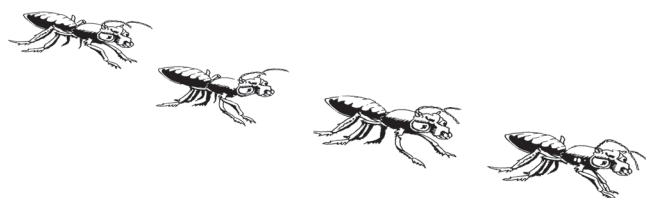


- 2 Ants had been coming into the house for the last few days, especially into the kitchen.



- 3 Jarrah had read that when a cat's eyes broaden, it's a sure sign of rain. When he had a look at his cat's eyes they appeared to be a little bit broader than normal.

- 4 On one occasion, Jarrah's neighbour's dog appeared to be eating grass, which he'd read was a sure sign of rain, but was the dog eating something on the grass? By the time he was able to get a closer look, the dog had moved to the garden bed and was digging up plants.



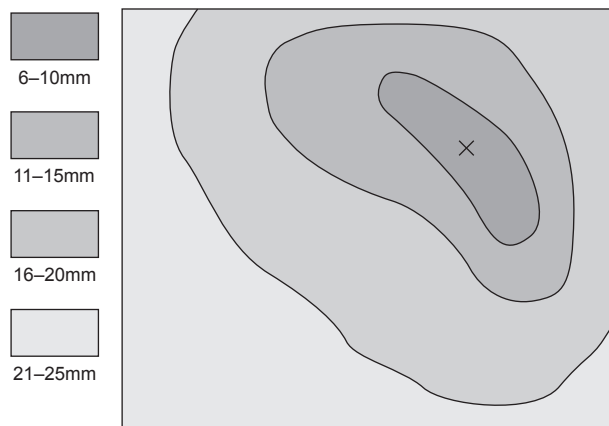
- 5 He thought he saw a bat flying early in the evening, which was a sign of good weather. Unless it was a raven, in which case it could mean rain.



- 6 Jarrah's family has a pond in which are several frogs. One was definitely shiny-skinned which meant fine weather was on the way. He hoped he didn't have any dull-skinned frogs, because that meant rain. Frog spawn around the edge of the ponds meant storms, so he cleared the edges just in case.



- 7 The moon seemed quite clear of any rings, and sunrises during the past few days had been a beautiful pale pink colour. Jarrah wondered whether that constituted the type of red mentioned in the well known saying: *Red sky in the morning, Sailor's warning!*



Initially fine and mild with the probability of a cool change and some rain by Sunday.

Monday: Mainly fine and sunny with light cloud and a moderate southerly wind. Max. temp. 23°C

Tuesday: Mostly fine and sunny with light cloud and a moderate southerly wind.
Max. temp. 25°C

Wednesday: Morning cloud clearing to a warm sunny day. Moderate northerly winds. Max. temp. 28°C

Thursday: A mostly warm sunny day. Light cloud and moderate northerly winds. Max. temp. 32°C

Friday: A sunny day with fresh northerly winds shifting to southerly by early evening. Max. temp. 36°C

Saturday: Mainly fine with a brief shower or two in the early morning. Light south to southwest winds.
Max. temp. 23°C

Sunday: Rain and strong southerly winds with the likelihood of a late thunderstorm. Max. temp. 17°C

2004 Below average seasonal falls

2003 Above average seasonal falls but overall a dry year

2002 Average seasonal rainfall

2001 Good seasonal rainfall

2000 Above average seasonal falls

2005 Warm lead-up to summer

2004 Average seasonal temperatures

2003 Above average seasonal temperatures

2002 Warm lead-up to summer

2001 Late spring warm spell

2000 Above average warm season

A La Niña weather pattern appears to be developing in the eastern Pacific which may bring above average rainfalls to Australia, early in the year. However, a drought-causing El Niño pattern may still occur later in the year.

The party

Write on the back of this page if you need more space.



**What is your recommendation to Jarrah?
Should the party be held indoors or outdoors?**

Reasons for your decision

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Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

